

SMAI 2025 spring (Jan-Aprl) PYQs

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16 – Quiz 2
38 – MidSem
49 – EndSem

4 sets for Quiz 1

Exam: 15 marks

Quiz 1

Set 0

Name:

Roll Number.....

- Answer precisely in the space given. No overwriting. Make assumptions, if really ambiguous.
- Questions 1-10: 1 Points; Questions 11-12: 2.5 Points. Total 15 Points. o

1. Consider a simple linear classifier, with bias. $\mathbf{w} = [2.5, 0.5, 1.5]^T$. This is tested on the following 5 samples. $(\mathbf{x}_i, y_i)$

$$([0, 0]^T, +), ([+1, +1]^T, +), ([-1, +1]^T, +), ([-1, -1]^T, -), ([+1, -1]^T, -)$$

What is the accuracy of this classifier ? Ans: _____-%.

2. A 3NN classifier is built over four training examples, and tested on the same test set of Q1.

$$([+2.0, 0.5]^T, +), (-3.0, +0.5]^T, -), ([0.0, 3.0]^T, +), ([0.0, -3.0]^T, -)$$

What is the accuracy of this classifier ? Ans: _____-%.

3. Posterior probability of two classes (i.e., $P(\omega_1|\mathbf{x})$ and $P(\omega_2|\mathbf{x})$) for two classes in a binary classification problem (with $\mathbf{x} \in \mathbb{R}^1$) are known to be normal distributions $\mathcal{N}(20, 25)$ and $\mathcal{N}(30, 5)$ respectively. Given a test sample of 25.1, predict the class. Ans: _____
4. Given the following confusion matrix.

	Predicted +ve	Predicted -ve
Actual +ve	A	B
Actual -ve	C	D

Write an expression for precision. Ans _____

Write an expression for accuracy. Ans _____

5. An $N \times N$ matrix A is composed of consecutive N^2 integers starting from K . (i.e., $K, K + 1, \dots, K + (N^2 - 1)$)

Rank of A is independent of N . (True or False?) _____

Rank of A is independent of K . (True or False?) _____

6. Bag I contain 10 white and 5 black balls. Bag II contains 15 white and 5 black balls.

A ball is drawn at random from one of the bags, and it is found to be white. What is the probability that it was drawn from Bag I.

Ans _____

7. A set of samples were pre-processed by a simple linear transformation $\mathbf{x}' = \mathbf{A}\mathbf{x}$ Let d_{ij} is the distance between $\mathbf{x}_i$ and $\mathbf{x}_j$ and d'_{ij} is the distance between $\mathbf{x}'_i$ and $\mathbf{x}'_j$

(a) When $\mathbf{A}$ is a permutation matrix (i.e., every row and column has only one '1' and all other elements being '0'; Note: $\mathbf{A}$ need not be identity). Then, $d_{ij} = d'_{ij}$ for all i, j . True or False? _____

(b) When $\mathbf{A}$ is $\rho\mathbf{I}$, a simple linear classifier $\text{sign}(\mathbf{w}^T \mathbf{x})$ will not report any change in accuracy after the transformation. True or False? _____

8. Consider a vocabulary of size d . One hot representation of a word i , $\mathbf{w}_i$, is “1” at the location (index) corresponding to that word and zero else where. Given a document that contains P words, $\mathbf{w}_1, \dots, \mathbf{w}_P$, we compute

$$\mathbf{x} = \sum_{i=1}^P \mathbf{w}_i$$

Then,

- (a) $\mathbf{x}$ is the histogram of the words, with its i th element x_i as the frequency of i th word.
- (b) $\mathbf{x}$ is in R^d independent of the number of words in the document.
- (c) $\mathbf{x}$ is in R^P independent of the vocabulary size.
- (d) $\sum_i x_i$ is P (x_i is the i th element of $\mathbf{x}$)

Which of the above statements are True? _____

9. Consider the covariance matrix Σ

- (a) Σ is symmetric
- (b) Σ is PSD
- (c) Σ is Diagonal if the distribution is Normal.
- (d) Σ can not be Diagonal if the distribution is Normal.
- (e) None of the above are true

Which of the above statements are True? _____

10. Consider the following statements:

- (a) Product of Eigen values is Determinant of a matrix
- (b) A matrix of $m \times n$ can have max (m,n) non-zero eigen values
- (c) If determinant of a matrix is zero, means at least one of the eigen value is zero.
- (d) Eigen vectors are orthogonal to each other (i.e., $\mathbf{v}_1^T \mathbf{v}_2 = 0$)
- (e) All the above are true

Which of the above statements are True? _____

11. We saw the loss function for linear regression as $J(\theta) = (Y - X\theta)^T(Y - X\theta)$. We saw that we get a closed form solution for θ by solving $\frac{\partial J(\theta)}{\partial \theta} = 0$:

$$\frac{\partial}{\partial \theta} (Y^T Y - 2Y^T X\theta + \theta^T X^T X\theta) = 0; \implies -2X^T Y + 2X^T X\theta = 0; \implies \theta = (X^T X)^{-1} X^T Y$$

Now find the closed form solution that minimizes this loss function (assume A is symmetric):

$$J(\theta) = (Y - X\theta)^T A(Y - X\theta)$$

- (a) $\theta = (X^T A X)^{-1} X^T Y$
- (b) $\theta = (X^T X)^{-1} X^T A Y$
- (c) $\theta = (X^T A X)^{-1} X^T A Y$
- (d) $\theta = (X^T A X)^{-1} X^T A^{-1} Y$
- (e) None of these

Write all correct options _____.

12. Consider the function

$$f(w) = w^2 + w + 1$$

We want to find the minima of the function using gradient descent. We start at $w^0 = 5.0$.

Write update equation for computing w^{k+1} from w^k . Ans: _____

What should be the learning rate η so that we reach the minima in a single step?

Ans: _____

Name:

Roll Number.....

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- Questions 1-10: 1 Points; Questions 11-12: 2.5 Points. Total 15 Points. 7

1. Consider a simple linear classifier, with bias. $\mathbf{w} = [2.5, 0.5, 1.5]^T$. This is tested on the following 5 samples. $(\mathbf{x}_i, y_i)$

$$([0, 0]^T, +), ([+1, +1]^T, +), ([-1, +1]^T, +), ([-1, -1]^T, -), ([+1, -1]^T, -)$$

How many errors this classifier makes? Ans: _____.

2. A 3NN classifier is built over four training examples, and tested on the same test set of Q1.

$$([+2.0, 0.5]^T, +), (-3.0, +0.5]^T, -), ([0.0, 3.0]^T, +), ([0.0, -3.0]^T, -)$$

What is the accuracy of this classifier ? Ans: _____%.

3. Posterior probability of two classes (i.e., $P(\omega_1|\mathbf{x})$ and $P(\omega_2|\mathbf{x})$) for two classes in a binary classification problem (with $\mathbf{x} \in R^1$) are known to be normal distributions $\mathcal{N}(30, 25)$ and $\mathcal{N}(40, 5)$ respectively. Given a test sample of 35.1, predict the class. Ans: _____
4. Given the following confusion matrix.

	Predicted +ve	Predicted -ve
Actual +ve	A	B
Actual -ve	C	D

Write an expression for recall. Ans: _____

Write an expression for accuracy. Ans: _____

5. An $N \times N$ matrix A is composed of consecutive N^2 integers starting from K . (i.e., $K, K + 1, \dots, K + (N^2 - 1)$)

Rank of A is independent of N . (True or False?) _____

Rank of A is independent of K . (True or False?) _____

6. Bag I contain 5 white and 10 black balls. Bag II contains 5 white and 15 black balls.

A ball is drawn at random from one of the bags, and it is found to be white. What is the probability that it was drawn from Bag I.

Ans: _____

7. A set of samples were pre-processed by a simple linear transformation $\mathbf{x}' = \mathbf{A}\mathbf{x}$ Let d_{ij} is the distance between $\mathbf{x}_i$ and $\mathbf{x}_j$ and d'_{ij} is the distance between $\mathbf{x}'_i$ and $\mathbf{x}'_j$

- (a) When $\mathbf{A}$ is a permutation matrix (i.e., every row and column has only one '1' and all other elements being '0'; Note: $\mathbf{A}$ need not be identity). Then, $d_{ij} = d'_{ij}$ for all i, j . True or False? _____

- (b) When $\mathbf{A}$ is $\rho \mathbf{I}$, a simple linear classifier $\text{sign}(\mathbf{w}^T \mathbf{x})$ will not report any change in accuracy after the transformation. True or False? ———
8. Consider a document is represented by a histogram of the words in the document. $\mathbf{h}$ i.e., h_i is the number of occurrence of the i th word in the document.

We define a linguistic operation: Paraphrasing (P1). P1 is defined as permuting sentences in a document and rewriting a sentence by permuting the words.

- (a) $\mathbf{h}$ is invariant to the P1
- (b) $\mathbf{h}$ is not invariant to the P1
- (c) $\mathbf{h}$ is invariant under in which order the vocabulary is constructed (eg. “a to z” or “z to a”)
- (d) a Euclidean distance computed over $\mathbf{h}_i$ and $\mathbf{h}_j$ is invariant under in which order the vocabulary is constructed (eg. “a to z” or “z to a”.)

Which of the above statements are True? ———

9. Consider the covariance matrix Σ

- (a) Σ is symmetric
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Which of the above statements are True? ———

11. We saw the loss function for linear regression as $J(\theta) = (Y - X\theta)^T(Y - X\theta)$. We saw that we get a closed form solution for θ by solving $\frac{\partial J(\theta)}{\partial \theta} = 0$:

$$\frac{\partial}{\partial \theta} (Y^T Y - 2Y^T X\theta + \theta^T X^T X\theta) = 0; \implies -2X^T Y + 2X^T X\theta = 0; \implies \theta = (X^T X)^{-1} X^T Y$$

Now find the closed form solution that minimizes this loss function (assume A is symmetric):

$$J(\theta) = (Y - X\theta)^T A (Y - X\theta)$$

- (a) $\theta = (X^T A X)^{-1} X^T Y$
- (b) $\theta = (X^T X)^{-1} X^T A Y$
- (c) $\theta = (X^T A X)^{-1} X^T A Y$
- (d) $\theta = (X^T A X)^{-1} X^T A^{-1} Y$
- (e) None of these

Write all correct options _____.

12. Consider the function

$$f(w) = w^2 + w + 1$$

We want to find the minima of the function using gradient descent. We start at $w^0 = -5.0$.

Write update equation for computing w^{k+1} from w^k . Ans: _____

What should be the learning rate η so that we reach the minima in a single step?

Ans: _____

Name:

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1. Consider a simple linear classifier, with bias. $\mathbf{w} = [2.5, 0.5, 1.5]^T$. This is tested on the following 5 samples. $(\mathbf{x}_i, y_i)$

$$([0, 0]^T, +), ([+1, +1]^T, +), ([-1, +1]^T, +), ([-1, -1]^T, -), ([+1, -1]^T, -)$$

What is the accuracy of this classifier ? Ans: _____%.

2. A 3NN classifier is built over four training examples, and tested on the same test set of Q1.

$$([+2.0, 0.5]^T, +), (-3.0, +0.5]^T, -), ([0.0, 3.0]^T, +), ([0.0, -3.0]^T, -)$$

How many errors this classifier makes ? Ans: _____.

3. Posterior probability of two classes (i.e., $P(\omega_1|\mathbf{x})$ and $P(\omega_2|\mathbf{x})$) for two classes in a binary classification problem (with $\mathbf{x} \in R^1$) are known to be normal distributions $\mathcal{N}(20, 5)$ and $\mathcal{N}(30, 25)$ respectively. Given a test sample of 24.9, predict the class. Ans: _____
4. Given the following confusion matrix.

	Predicted +ve	Predicted -ve
Actual +ve	A	B
Actual -ve	C	D

Write an expression for accuracy. Ans _____

Write an expression for precision. Ans _____

5. An $N \times N$ matrix A is composed of consecutive N^2 integers starting from K . (i.e., $K, K+1, \dots, K+(N^2-1)$)

Rank of A is independent of N . (True or False?) _____

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A ball is drawn at random from one of the bags, and it is found to be white. What is the probability that it was drawn from Bag I.

Ans _____

7. A set of samples were pre-processed by a simple linear transformation $\mathbf{x}' = \mathbf{A}\mathbf{x}$. Let d_{ij} is the distance between $\mathbf{x}_i$ and $\mathbf{x}_j$ and d'_{ij} is the distance between $\mathbf{x}'_i$ and $\mathbf{x}'_j$

(a) When $\mathbf{A}$ is a permutation matrix (i.e., every row and column has only one '1' and all other elements being '0'; Note: $\mathbf{A}$ need not be identity). Then, $d_{ij} = d'_{ij}$ for all i, j . True or False? _____

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8. Consider a document is represented by a histogram of the words in the document $\mathbf{h}$ i.e., h_i is the number of occurrence of the i th word in the document.

We define a linguistic operation: Paraphrasing (P2). P2 is defined as replacing a set of words by their respective synonym.

- (a) $\mathbf{h}$ is invariant to the P2
- (b) $\mathbf{h}$ is not invariant to the P2
- (c) $\mathbf{h}$ is invariant under in which order the vocabulary is constructed (eg. “a to z” or “z to a”)
- (d) a Euclidean distance computed over $\mathbf{h}_i$ and $\mathbf{h}_j$ is invariant under in which order the vocabulary is constructed (eg. “a to z” or “z to a”)

Which of the above statements are True? _____

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We want to find the minima of the function using gradient descent. We start at $w^0 = 5.0$.

Write update equation for computing w^{k+1} from w^k . Ans: _____

What should be w^1 if the learning rate $\eta = 0.1$?

Ans: _____

Name:

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1. Consider a simple linear classifier, with bias. $\mathbf{w} = [2.5, 0.5, 1.5]^T$. This is tested on the following 5 samples. $(\mathbf{x}_i, y_i)$

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4. Given the following confusion matrix.

	Predicted +ve	Predicted -ve
Actual +ve	A	B
Actual -ve	C	D

Write an expression for accuracy. Ans: _____

Write an expression for recall. Ans: _____

5. An $N \times N$ matrix A is composed of consecutive N^2 integers starting from K . (i.e., $K, K + 1, \dots, K + (N^2 - 1)$)

Rank of A is independent of N . (True or False?) _____

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A ball is drawn at random from one of the bags, and it is found to be white. What is the probability that it was drawn from Bag I. Ans: _____

7. A set of samples were pre-processed by a simple linear transformation $\mathbf{x}' = \mathbf{A}\mathbf{x}$ Let d_{ij} is the distance between $\mathbf{x}_i$ and $\mathbf{x}_j$ and d'_{ij} is the distance between $\mathbf{x}'_i$ and $\mathbf{x}'_j$

(a) When $\mathbf{A}$ is a permutation matrix (i.e., every row and column has only one '1' and all other elements being '0'; Note: $\mathbf{A}$ need not be identity). Then, $d_{ij} = d'_{ij}$ for all i, j . True or False? _____

(b) When $\mathbf{A}$ is $\rho\mathbf{I}$, a simple linear classifier $\text{sign}(\mathbf{w}^T \mathbf{x})$ will not report any change in accuracy after the transformation. True or False? _____

8. Consider a vocabulary of size d . One hot representation of a word i , $\mathbf{w}_i$, is “1” at the location (index) corresponding to that word and zero else where. Given a document that contains P words, $\mathbf{w}_1, \dots, \mathbf{w}_P$, we compute

$$\mathbf{x} = \sum_{i=1}^P \mathbf{w}_i$$

Then,

- (a) $\mathbf{x}$ is the histogram of the words, with its i th element x_i as the frequency of i th word.
- (b) $\mathbf{x}$ is in R^d independent of the number of words in the document.
- (c) $\mathbf{x}$ is in R^P independent of the vocabulary size.
- (d) $\sum_i x_i$ is P (x_i is the i th element of $\mathbf{x}$)

Which of the above statements are True? _____

9. Consider the covariance matrix Σ

- (a) Σ is symmetric
- (b) Σ is PSD
- (c) Σ is Diagonal if the distribution is Normal.
- (d) Σ can not be Diagonal if the distribution is Normal.
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Which of the above statements are True? _____

11. We saw the loss function for linear regression as $J(\theta) = (Y - X\theta)^T(Y - X\theta)$. We saw that we get a closed form solution for θ by solving $\frac{\partial J(\theta)}{\partial \theta} = 0$:

$$\frac{\partial}{\partial \theta} (Y^T Y - 2Y^T X\theta + \theta^T X^T X\theta) = 0; \implies -2X^T Y + 2X^T X\theta = 0; \implies \theta = (X^T X)^{-1} X^T Y$$

Now find the closed form solution that minimizes this loss function (assume A is symmetric):

$$J(\theta) = (Y - X\theta)^T A (Y - X\theta)$$

- (a) $\theta = (X^T A X)^{-1} X^T Y$
- (b) $\theta = (X^T X)^{-1} X^T A Y$
- (c) $\theta = (X^T A X)^{-1} X^T A Y$
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- (e) None of these

Write all correct options _____.

12. Consider the function

$$f(w) = w^2 + w + 1$$

We want to find the minima of the function using gradient descent. We start at $w^0 = -5.0$.

Write update equation for computing w^{k+1} from w^k . Ans: _____

What should be w^1 if the learning rate $\eta = 0.1$?

Ans: _____

set $\neq$

1 $\rightarrow$ 2

2 $\rightarrow$ 80%

3 $\rightarrow$ class 1 | ω_1

4 $\rightarrow$ Recall = $\frac{A}{A+B}$ | Acc = $\frac{A+D}{A+B+C+D}$

5 $\rightarrow$ false, True

6 $\rightarrow$ $\frac{4}{7}$

7 $\rightarrow$ True, False

8 $\rightarrow$ A, D

9 $\rightarrow$ A, B

10 $\rightarrow$ A, C

11 $\rightarrow$ C

12 $\rightarrow$ $\omega^{k+1} = \omega^k - \eta (2\omega^k + 1)$
 $\eta = 0.5$

set {Empty}

1 $\rightarrow$ 2

2 $\rightarrow$ 1

3 $\rightarrow$ ω_2 | class 2

4 $\rightarrow$ Acc = $\frac{A+D}{A+B+C+D}$ || Recall = $\frac{A}{A+B}$

5 $\rightarrow$ false, True

6 $\rightarrow$ $\frac{3}{7}$

7 $\rightarrow$ True, false

8 $\rightarrow$ A, B, D

9 $\rightarrow$ A, B

10 $\rightarrow$ A, C

11 $\rightarrow$ C

12 $\rightarrow$ $\omega^{k+1} = \omega^k - \eta \cdot (2\omega^k + 1)$

$\omega^1 = -4.2$

set $\neq$

1 $\rightarrow$ 60%

2 $\rightarrow$ 1

3 $\rightarrow$ ω_2 (class 2)

4 $\rightarrow$ Acc = $\frac{A+D}{A+B+C+D}$ | precision = $\frac{A}{A+C}$

5 $\rightarrow$ False, True

6 $\rightarrow$ $\frac{9}{17}$

7 $\rightarrow$ True, False

8 $\rightarrow$ B

9 $\rightarrow$ A, B

10 $\rightarrow$ A, C

11 $\rightarrow$ C

12 $\rightarrow$ $\omega^{k+1} = \omega^k - \eta (2\omega^k + 1)$

$\omega^1 = 3.9$

set 0

1 $\rightarrow$ 60%

2 $\rightarrow$ 80%

3 $\rightarrow$ class 3 | ω_1

4 $\rightarrow$ Precision $\rightarrow \frac{A}{A+C}$ | Acc = $\frac{A+D}{A+B+C+D}$

5 $\rightarrow$ False, True

6 $\rightarrow$ $\frac{6}{17}$

7 $\rightarrow$ True, False

8 $\rightarrow$ A, B, D

9 $\rightarrow$ A, B

10 $\rightarrow$ A, C

11 $\rightarrow$ C

12 $\rightarrow$ $\omega^{k+1} = \omega^k - \eta (2\omega^k + 1)$
 $\eta = 0.5$

Quiz 2

40 marks , 8 ques

$$\rightarrow 8 \times 5 = 40$$

Quiz II

Name:

Roll Number.....

-
- Answer precisely in the space given. No overwriting. Make assumptions, if really ambiguous.
 - Do rough work in the additional sheet.
-

Fill in the blank with precise answers.

[8 × 5 = 40]

1. Consider a simple neural network with two inputs, one hidden layer with three neurons and one output. No bias. If the input is $x_1 = 3$ and $x_2 = 2$, output is ——— (fact: All weights are 1.0. Activation functions in the hidden layer is ReLU and activation function in the output is linear.)

Solution:-

The neural network processes the inputs as follows:

1. Hidden Layer Calculation:

- Each hidden neuron receives inputs $x_1 = 3$ and $x_2 = 2$ with weights of 1.0.
- Pre-activation: $(3 \times 1) + (2 \times 1) = 5$.
- ReLU activation: $\text{ReLU}(5) = 5$.
- All three hidden neurons output 5.

2. Output Layer Calculation:

- The output neuron sums the three hidden layer outputs (each multiplied by weight 1.0):

$$5 \times 1 + 5 \times 1 + 5 \times 1 = 15.$$

- Linear activation passes the sum directly.

Output: 15

2. Continuing the about question, assume the target (t) of the above sample was 25, and the loss was squared error, $\frac{\partial \mathcal{L}}{\partial w_{11}}$ is ——— (w_{11} is the weight from first input (i.e., x_1) to first neuron in the hidden layer.)

Solution:-

To compute $\frac{\partial \mathcal{L}}{\partial w_{11}}$ using backpropagation:

1. Loss Derivative:

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = 2(\hat{y} - t) = 2(15 - 25) = -20.$$

2. Output Layer Gradient:

$$\frac{\partial \hat{y}}{\partial h_1} = 1 \quad (\text{since } \hat{y} = h_1 + h_2 + h_3).$$

3. Hidden Layer Activation Gradient:

$$\frac{\partial h_1}{\partial z_1} = 1 \quad (\text{ReLU derivative at } z_1 = 5).$$

4. Pre-activation Gradient:

$$\frac{\partial z_1}{\partial w_{11}} = x_1 = 3.$$

Chain Rule:

$$\frac{\partial \mathcal{L}}{\partial w_{11}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{11}} = (-20) \cdot 1 \cdot 1 \cdot 3 = -60$$

Answer: -60

3. Consider a single layer perceptron with two input and one output. The weights from first and second inputs are w_1 and w_2 respectively. Bias is w_0 . Also assume a ± 1 logic. The activation at the output is: $\phi(x) = +1$ if $x \geq 0$ and -1 else. If $w_0 = 1, w_1 = -1, w_2 = -1$, then this neuron model is equivalent to: _____ (fill from the gates like: AND, OR, ExOR, NAND, NOR etc.)

Solution:-

Consider the perceptron with two inputs and one output. The net input to the neuron is given by:

$$x = w_0 \cdot 1 + w_1 x_1 + w_2 x_2$$

Given that $w_0 = 1, w_1 = -1$, and $w_2 = -1$, we compute x for all possible input values x_1, x_2 (where $x_1, x_2 \in \{+1, -1\}$):

$$x = 1 - x_1 - x_2$$

The activation function is:

$$\phi(x) = \begin{cases} +1, & \text{if } x \geq 0 \\ -1, & \text{otherwise} \end{cases}$$

We now evaluate the truth table:

x_1	x_2	$x = 1 - x_1 - x_2$	$\phi(x)$
+1	+1	$1 - 1 - 1 = -1$	-1
+1	-1	$1 - 1 - (-1) = 1$	+1
-1	+1	$1 - (-1) - 1 = 1$	+1
-1	-1	$1 - (-1) - (-1) = 3$	+1

This matches the truth table of the **NAND** gate (in ± 1 logic):

$$\text{NAND}(x_1, x_2) = \begin{cases} +1, & \text{if } (x_1, x_2) \neq (+1, +1) \\ -1, & \text{if } (x_1, x_2) = (+1, +1) \end{cases}$$

Thus, this perceptron implements the **NAND** gate.

4. Consider a binary classification problem (A vs B) with perceptron and bias. Consider two samples from A: $[1, 2]^T, [4, 2]^T$ and one sample from B: $[3, 0]^T$. We initialize $\mathbf{w}^0 = [1, 1, 0]^T$. When learning rate $\eta = 0.1$, $\mathbf{w}^1$ (i.e., after one iteration of perceptron algorithm) is _____

Solution:-

We are given:

- Class A samples (label +1): $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$
- Class B sample (label -1): $\mathbf{x}_3 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$
- Initial weights: $\mathbf{w}^{(0)} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$
- Learning rate: $\eta = 0.1$

We augment each input vector with a bias term of 1:

$$\tilde{\mathbf{x}}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \tilde{\mathbf{x}}_2 = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}, \quad \tilde{\mathbf{x}}_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

Now we apply one iteration of the Perceptron algorithm:

- For $\tilde{\mathbf{x}}_1, y = +1$:
 $\mathbf{w}^{(0)} \cdot \tilde{\mathbf{x}}_1 = [1, 1, 0] \cdot [1, 2, 1] = 3 > 0 \Rightarrow \text{Correct}$
- For $\tilde{\mathbf{x}}_2, y = +1$:
 $\mathbf{w}^{(0)} \cdot \tilde{\mathbf{x}}_2 = [1, 1, 0] \cdot [4, 2, 1] = 6 > 0 \Rightarrow \text{Correct}$
- For $\tilde{\mathbf{x}}_3, y = -1$:
 $\mathbf{w}^{(0)} \cdot \tilde{\mathbf{x}}_3 = [1, 1, 0] \cdot [3, 0, 1] = 3 > 0 \Rightarrow \text{Incorrect}$

Update rule:

$$\mathbf{w}^{(1)} = \mathbf{w}^{(0)} + \eta \cdot y \cdot \tilde{\mathbf{x}}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + 0.1 \cdot (-1) \cdot \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 1 \\ -0.1 \end{bmatrix}$$

Final answer:

$$\boxed{\mathbf{w}^{(1)} = \begin{bmatrix} 0.7 \\ 1 \\ -0.1 \end{bmatrix}}$$

5. Consider a 1D binary classification problem (2.5,+), (4.2,+), (5.6,-), (7.8,-). We train a hard margin SVM. The support vector(s) is/are _____ (if there are more than one SV, write all.)

Solution:-

We are given the one-dimensional dataset:

$$\{(2.5, +), (4.2, +), (5.6, -), (7.8, -)\}.$$

For a hard-margin SVM with decision function

$$f(x) = wx + b,$$

we must satisfy the constraint

$$y_i(wx_i + b) \geq 1 \quad \text{for all } i.$$

Analysis:

- The positive class consists of points at $x = 2.5$ and $x = 4.2$
- The negative class consists of points at $x = 5.6$ and $x = 7.8$
- The decision boundary should lie between the rightmost positive point and leftmost negative point

Identifying Support Vectors:

- The rightmost positive point: 4.2
- The leftmost negative point: 5.6
- These are the closest points between the two classes and will satisfy:

$$w \cdot 4.2 + b = +1 \quad \text{and} \quad w \cdot 5.6 + b = -1.$$

- Solving these gives the optimal hyperplane parameters

Verification:

- Point at 2.5: $w \cdot 2.5 + b \geq 1$ (satisfied with margin)
- Point at 7.8: $w \cdot 7.8 + b \leq -1$ (satisfied with margin)
- Only 4.2 and 5.6 lie exactly on the margin boundaries

Thus, the support vectors that define the optimal margin are:

$$\boxed{(4.2, +) \quad \text{and} \quad (5.6, -)}.$$

6. Consider the following dataset $\mathcal{D}$ of 5 samples in 2D, $\mathcal{D} = \{[1, 1]^T, [2, 2]^T, [3, 3]^T, [4, 4]^T, [5, 5]^T\}$.
The covariance of $\mathcal{D}$ is _____

Solution:-

Given the 2D dataset:

$$D = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ 5 \end{bmatrix} \right\}$$

Step 1: Compute the Mean Vector

$$\mu = \frac{1}{5} \sum_{i=1}^5 \mathbf{x}_i = \frac{1}{5} \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 4 \end{bmatrix} + \begin{bmatrix} 5 \\ 5 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

Step 2: Center the Data Subtract the mean from each data point:

$$\mathbf{x}'_i = \mathbf{x}_i - \mu$$

The centered data becomes:

$$\left\{ \begin{bmatrix} -2 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}$$

Step 3: Compute the Covariance Matrix For a 2D dataset, the covariance matrix is:

$$\Sigma = \frac{1}{n} \sum_{i=1}^n \mathbf{x}'_i \mathbf{x}'_i{}^T$$

Calculating each term:

$$\begin{aligned} & \frac{1}{5} \left(\begin{bmatrix} -2 \\ -2 \end{bmatrix} \begin{bmatrix} -2 & -2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} \begin{bmatrix} -1 & -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \end{bmatrix} \right) \\ &= \frac{1}{5} \left(\begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \right) \\ &= \frac{1}{5} \begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \end{aligned}$$

Final Answer: The covariance matrix of dataset D is:

$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Interpretation: The identical diagonal elements (2) indicate equal variance in both dimensions, and the off-diagonal elements (2) show perfect positive correlation between the two dimensions, as all points lie exactly on the line $y=x$.

7. A dataset $\mathcal{D}$ consists of six feature measurements. $\mathcal{D} = \{2.2, 3.5, 4.8, 1.7, 2.8, 3.0\}$. $\mathcal{D}$ needs to be normalized such that mean is zero and variance is unity. The normalized $\mathcal{D}$ is _____

Solution:-

Given the dataset:

$$D = \{2.2, 3.5, 4.8, 1.7, 2.8, 3.0\}$$

Step 1: Compute the Mean (μ)

$$\mu = \frac{1}{6} \sum_{i=1}^6 x_i = \frac{2.2 + 3.5 + 4.8 + 1.7 + 2.8 + 3.0}{6} = \frac{18.0}{6} = 3.0$$

Step 2: Compute the Variance (σ^2) First, calculate the squared differences from the mean:

$$(2.2 - 3.0)^2 = 0.64$$

$$(3.5 - 3.0)^2 = 0.25$$

$$(4.8 - 3.0)^2 = 3.24$$

$$(1.7 - 3.0)^2 = 1.69$$

$$(2.8 - 3.0)^2 = 0.04$$

$$(3.0 - 3.0)^2 = 0.00$$

$$\sigma^2 = \frac{0.64 + 0.25 + 3.24 + 1.69 + 0.04 + 0.00}{6} = \frac{5.86}{6} \approx 0.9767$$

$$\sigma = \sqrt{0.9767} \approx 0.9883$$

Step 3: Normalize Each Data Point Using the formula:

$$z_i = \frac{x_i - \mu}{\sigma}$$

$$z_1 = \frac{2.2 - 3.0}{0.9883} \approx -0.809$$

$$z_2 = \frac{3.5 - 3.0}{0.9883} \approx 0.506$$

$$z_3 = \frac{4.8 - 3.0}{0.9883} \approx 1.822$$

$$z_4 = \frac{1.7 - 3.0}{0.9883} \approx -1.316$$

$$z_5 = \frac{2.8 - 3.0}{0.9883} \approx -0.202$$

$$z_6 = \frac{3.0 - 3.0}{0.9883} = 0.000$$

Final Answer: The normalized dataset is:

$$\{-0.809, 0.506, 1.822, -1.316, -0.202, 0.000\}$$

Verification:

- Mean of normalized data: $(-0.809 + 0.506 + 1.822 - 1.316 - 0.202 + 0.000)/6 \approx 0$
- Variance of normalized data: ≈ 1 (by construction)

8. We know for a valid kernel $\kappa(\mathbf{p}, \mathbf{q})$, there exists a $\phi(\mathbf{p})^T \phi(\mathbf{q})$ form where ϕ does an appropriate feature transformation on $\mathbf{p}$ and $\mathbf{q}$. When $\mathbf{p} = [1, 1]^T$; $\mathbf{q} = [2, 2]^T$ and $\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q} + 1)^2$, find the corresponding ϕ and write as $\phi(\mathbf{p})^T \phi(\mathbf{q})$ ————— (such as $[- - - - -]^T [- - - - -]$)

Solution:-

Given the kernel function:

$$\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q} + 1)^2$$

with input vectors:

$$\mathbf{p} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Step 1: Expand the Kernel Function

$$\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q} + 1)^2 = (1 \cdot 2 + 1 \cdot 2 + 1)^2 = (2 + 2 + 1)^2 = 25$$

Step 2: Identify the Feature Mapping The polynomial kernel $(x^T y + 1)^2$ corresponds to the feature mapping:

$$\phi(\mathbf{x}) = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ 1 \end{bmatrix}$$

Step 3: Compute $\phi(\mathbf{p})$ and $\phi(\mathbf{q})$ For $\mathbf{p} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$\phi(\mathbf{p}) = \begin{bmatrix} 1^2 \\ 1^2 \\ \sqrt{2} \cdot 1 \cdot 1 \\ \sqrt{2} \cdot 1 \\ \sqrt{2} \cdot 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \\ \sqrt{2} \\ \sqrt{2} \\ 1 \end{bmatrix}$$

For $\mathbf{q} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$:

$$\phi(\mathbf{q}) = \begin{bmatrix} 2^2 \\ 2^2 \\ \sqrt{2} \cdot 2 \cdot 2 \\ \sqrt{2} \cdot 2 \\ \sqrt{2} \cdot 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4\sqrt{2} \\ 2\sqrt{2} \\ 2\sqrt{2} \\ 1 \end{bmatrix}$$

Step 4: Verify $\phi(\mathbf{p})^T \phi(\mathbf{q}) = (p, q)\phi(\mathbf{p})^T \phi(\mathbf{q}) = 1 \cdot 4 + 1 \cdot 4 + \sqrt{2} \cdot 4\sqrt{2} + \sqrt{2} \cdot 2\sqrt{2} + \sqrt{2} \cdot 2\sqrt{2} + 1 \cdot 1$
 $= 4 + 4 + 8 + 4 + 4 + 1 = 25$ (matches kernel calculation)

Final Answer: The feature transformation is:

$$\phi(\mathbf{p})^T \phi(\mathbf{q}) = \begin{bmatrix} 1 & 1 & \sqrt{2} & \sqrt{2} & \sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \\ 4\sqrt{2} \\ 2\sqrt{2} \\ 2\sqrt{2} \\ 1 \end{bmatrix}$$

Quiz II

Name:

Roll Number.....

- Answer precisely in the space given. No overwriting. Make assumptions, if really ambiguous.
- Do rough work in the additional sheet.

Fill in the blank with precise answers;

[8 × 5 = 40]

1. Consider a simple neural network with two inputs, one hidden layer with three neurons and one output. No bias. If the input is $x_1 = 3$ and $x_2 = 3$, output is ——— (fact: All weights are 1.0. Activation functions in the hidden layer is ReLU and activation function in the output is linear.)

Answer:

Each hidden neuron computes the following.

$$\begin{aligned} z_i &= w_{i1}x_1 + w_{i2}x_2 & i \in \{1, 2, 3\} \\ &= 3 + 3 \\ &= 6 \\ a_i &= \max(0, z_i) & i \in \{1, 2, 3\} \\ &= 6 \end{aligned}$$

The final output is a linear combination of the hidden neuron outputs with weights 1.

$$\begin{aligned} \hat{y} &= \sum_{i=1}^3 a_i \\ &= 6 + 6 + 6 \\ &= \boxed{18} \end{aligned}$$

2. Continuing the about question, assume the target (t) of the above sample was 25, and the loss was squared error, $\frac{\partial \mathcal{L}}{\partial w_{11}}$ is ——— (w_{11} is the weight from first input (i.e., x_1) to first neuron in the hidden layer.)

Answer:

We know $y = 25$, $\hat{y} = 18$. We can write the squared error as follows.

$$\mathcal{L} = (y - \hat{y})^2,$$

The gradient with respect to the final output can be computed as follows.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \hat{y}} &= -2(y - \hat{y}) \\ &= -2(25 - 18) \\ &= -14 \end{aligned}$$

It is easy to see that the following holds, when the output layer weights are tied to one.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial} &= \frac{\partial \mathcal{L}}{\partial a_i} & i \in \{1, 2, 3\} \\ &= \frac{\partial \mathcal{L}}{\partial z_i} & i \in \{1, 2, 3\} \end{aligned}$$

Finally, the gradient with respect to the linear layer weights can be computed as follows.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial w_{11}} &= \frac{\partial \mathcal{L}}{\partial z_i} \times \frac{\partial z_i}{\partial w_{11}} \\ &= -14 \times x_1 \\ &= -14 \times 3 \\ &= \boxed{-42} \end{aligned}$$

3. Consider a single layer perceptron with two input and one output. The weights from first and second inputs are w_1 and w_2 respectively. Bias is w_0 . Also assume a ± 1 logic. The activation at the output is: $\phi(x) = +1$ if $x \geq 0$ and -1 else. If $w_0 = 1, w_1 = -1, w_2 = -1$, then this neuron model is equivalent to: _____ (fill from the gates like: AND, OR, ExOR, NAND, NOR etc.)

Answer:

The output is computed as follows.

$$z = 1 - x_1 - x_2$$

$$\phi(x) = \begin{cases} +1, & \text{if } z \geq 0 \\ -1, & \text{if } z < 0 \end{cases}$$

Computing the truth table for all input combinations $(x_1, x_2) \in \{+1, -1\}^2$ shows the model resembles NAND gate.

x_1	x_2	z	$\phi(x)$
1	1	-1	-1
1	-1	1	1
-1	1	1	1
-1	-1	3	1

4. Consider a binary classification problem (A vs B) with perceptron and bias. Consider two samples from A: $[1, 2]^T, [4, 2]^T$ and one sample from B: $[3, 0]^T$. We initialize $\mathbf{w}^0 = [1, 1, 0]^T$. When learning rate $\eta = 0.1$, $\mathbf{w}^1$ (i.e., after one iteration of perceptron algorithm) is _____

Answer:

The output of the perceptron can be computed as follows. Note that the input has to be prepended with a 1 for accommodating the bias weight.

$$\hat{y}_i = w^T x_i$$

$$\hat{y}_1 = [1 \quad 1 \quad 0] \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 2$$

$$\hat{y}_2 = [1 \quad 1 \quad 0] \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} = 5$$

$$\hat{y}_3 = [1 \quad 1 \quad 0] \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} = 4$$

The initial weight can be updated for the misclassified sample 3 (of class B) as follows.

$$\begin{aligned} \mathbf{w}^1 &= \mathbf{w}^0 - \eta \mathbf{x}_3 \\ &= \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0.9 \\ 0.7 \\ 0 \end{bmatrix} \end{aligned}$$

Some students have assumed that the input is appended by 1, rather than prepended. The weight update in that case would be as follows.

$$\begin{aligned}\mathbf{w}^1 &= \mathbf{w}^0 - \eta \mathbf{x}_3 \\ &= \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.7 \\ 1 \\ -0.1 \end{bmatrix}\end{aligned}$$

5. Consider a 1D binary classification problem (2.5,+), (4.2,-), (5.6,-), (7.8,-). We train a hard margin SVM. The support vector(s) is/are _____ (if there are more than one SV, write all.)

Answer:

The hard margin SVM places the decision boundary midway between the lone positive and the closest negative. Thus the support vectors are at $\boxed{2.5, 4.2}$.

6. Consider the following dataset $\mathcal{D}$ of 5 samples in 2D, $\mathcal{D} = \{[1, 2]^T, [2, 1]^T, [3, 3]^T, [4, 4]^T, [5, 5]^T\}$. The covariance of $\mathcal{D}$ is _____

Answer:

The mean and variance of the sequence $\{1, 2, \dots, 5\}$ are 3 and 2 respectively (easy to see). The covariance can be computed as follows.

$$\text{Cov}(x, y) = \frac{2 + 2 + 0 + 1 + 4}{5} = 1.8$$

The covariance matrix can thus be written as follows.

$$\Sigma(\mathcal{D}) = \begin{bmatrix} 2 & 1.8 \\ 1.8 & 2 \end{bmatrix}$$

7. A dataset $\mathcal{D}$ consists of six feature measurements. $\mathcal{D} = \{3.2, 4.5, 5.8, 2.7, 3.8, 4.0\}$. $\mathcal{D}$ needs to be normalized such that mean is zero and variance is unity. The normalized $\mathcal{D}$ is _____

Answer:

The mean and variance can be computed as follows.

$$\mu = \frac{3.2 + 4.5 + 5.8 + 2.7 + 3.8 + 4.0}{6} = 4$$

$$\sigma^2 = \frac{(3.2 - 4)^2 + (4.5 - 4)^2 + (5.8 - 4)^2 + (2.7 - 4)^2 + (3.8 - 4)^2 + (4.0 - 4)^2}{6} \approx 0.98$$

Finally, the normalized dataset can be computed as follows.

$$\left\{ \frac{3.2 - 4}{0.99}, \frac{4.5 - 4}{0.99}, \frac{5.8 - 4}{0.99}, \frac{2.7 - 4}{0.99}, \frac{3.8 - 4}{0.99}, \frac{4.0 - 4}{0.99} \right\} \approx \boxed{\{-0.80, 0.51, 1.82, -1.31, -0.20, 0\}}$$

8. We know for a valid kernel $\kappa(\mathbf{p}, \mathbf{q})$, there exists a $\phi(\mathbf{p})^T \phi(\mathbf{q})$ form where ϕ does an appropriate feature transformation on $\mathbf{p}$ and $\mathbf{q}$. When $\mathbf{p} = [1, 2]^T$; $\mathbf{q} = [2, 1]^T$ and $\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q} + 1)^2$, find the corresponding ϕ and write as $\phi(\mathbf{p})^T \phi(\mathbf{q})$ _____ (such as $[- \text{---} - \text{---} - \text{---} - \text{---}]^T [- \text{---} - \text{---} - \text{---} - \text{---}]$)

Answer:

The kernel can be expanded as follows.

$$\begin{aligned}
\kappa(\mathbf{p}, \mathbf{q}) &= (\mathbf{p}^T \mathbf{q} + 1)^2 \\
&= (p_1 q_1 + p_2 q_2 + 1)^2 \\
&= p_1^2 q_1^2 + 2p_1 p_2 q_1 q_2 + p_2^2 q_2^2 + 2p_1 q_1 + 2p_2 q_2 + 1
\end{aligned}$$

We would like to separate the function $\kappa(\mathbf{p}, \mathbf{q})$ as $\phi(\mathbf{p})^T \phi(\mathbf{q})$. Since the function is symmetric in $\mathbf{p}$ and $\mathbf{q}$, this can be achieved by matching the terms in the sum as follows.

$$\phi(\mathbf{p}) = \begin{bmatrix} p_1^2 \\ \sqrt{2} p_1 p_2 \\ p_2^2 \\ \sqrt{2} p_1 \\ \sqrt{2} p_2 \\ 1 \end{bmatrix}$$

Note that since inner product is invariant to permutation, any permutation of the feature transformation would meet the requirement.

Quiz II

Name:

Roll Number.....

-
- Answer precisely in the space given. No overwriting. Make assumptions, if really ambiguous.
 - Do rough work in the additional sheet.
-

Fill in the blank with precise answers,

[8 × 5 = 40]

1. Consider a simple neural network with two inputs, one hidden layer with three neurons and one output. No bias. If the input is $x_1 = 2$ and $x_2 = 2$, output is ——— (fact: All weights are 1.0. Activation functions in the hidden layer is ReLU and activation function in the output is linear.)

Solution:-

The neural network processes the inputs as follows:

1. Hidden Layer Calculation:

- Each hidden neuron receives inputs $x_1 = 2$ and $x_2 = 2$ with weights of 1.0.
- Pre-activation: $(2 \times 1) + (2 \times 1) = 4$.
- ReLU activation: $\text{ReLU}(4) = 4$.
- All three hidden neurons output 4.

2. Output Layer Calculation:

- The output neuron sums the three hidden layer outputs (each multiplied by weight 1.0):

$$4 \times 1 + 4 \times 1 + 4 \times 1 = 12.$$

- Linear activation passes the sum directly.

Output: 12

2. Continuing the about question, assume the target (t) of the above sample was 25, and the loss was squared error, $\frac{\partial \mathcal{L}}{\partial w_{11}}$ is ——— (w_{11} is the weight from first input (i.e., x_1) to first neuron in the hidden layer.)

Solution:-

To compute $\frac{\partial \mathcal{L}}{\partial w_{11}}$ using backpropagation:

1. Loss Derivative:

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = 2(\hat{y} - t) = 2(12 - 25) = -26.$$

2. Output Layer Gradient:

$$\frac{\partial \hat{y}}{\partial h_1} = 1 \quad (\text{since } \hat{y} = h_1 + h_2 + h_3).$$

3. Hidden Layer Activation Gradient:

$$\frac{\partial h_1}{\partial z_1} = 1 \quad (\text{ReLU derivative at } z_1 = 4).$$

4. Pre-activation Gradient:

$$\frac{\partial z_1}{\partial w_{11}} = x_1 = 2.$$

Chain Rule:

$$\frac{\partial \mathcal{L}}{\partial w_{11}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{11}} = (-26) \cdot 1 \cdot 1 \cdot 2 = -52$$

Answer: -52

3. Consider a single layer perceptron with two input and one output. The weights from first and second inputs are w_1 and w_2 respectively. Bias is w_0 . Also assume a ± 1 logic. The activation at the output is: $\phi(x) = +1$ if $x \geq 0$ and -1 else. If $w_0 = 1, w_1 = -1, w_2 = -1$, then this neuron model is equivalent to: _____ (fill from the gates like: AND, OR, ExOR, NAND, NOR etc.)

Solution:-

Consider the perceptron with two inputs and one output. The net input to the neuron is given by:

$$x = w_0 \cdot 1 + w_1 x_1 + w_2 x_2$$

Given that $w_0 = 1, w_1 = -1$, and $w_2 = -1$, we compute x for all possible input values x_1, x_2 (where $x_1, x_2 \in \{+1, -1\}$):

$$x = 1 - x_1 - x_2$$

The activation function is:

$$\phi(x) = \begin{cases} +1, & \text{if } x \geq 0 \\ -1, & \text{otherwise} \end{cases}$$

We now evaluate the truth table:

x_1	x_2	$x = 1 - x_1 - x_2$	$\phi(x)$
+1	+1	$1 - 1 - 1 = -1$	-1
+1	-1	$1 - 1 - (-1) = 1$	+1
-1	+1	$1 - (-1) - 1 = 1$	+1
-1	-1	$1 - (-1) - (-1) = 3$	+1

This matches the truth table of the **NAND** gate (in ± 1 logic):

$$\text{NAND}(x_1, x_2) = \begin{cases} +1, & \text{if } (x_1, x_2) \neq (+1, +1) \\ -1, & \text{if } (x_1, x_2) = (+1, +1) \end{cases}$$

Thus, this perceptron implements the **NAND** gate.

4. Consider a binary classification problem (A vs B) with perceptron and bias. Consider two samples from A: $[1, 2]^T, [4, 2]^T$ and one sample from B: $[3, 0]^T$. We initialize $\mathbf{w}^0 = [1, 1, 0]^T$. When learning rate $\eta = 0.1$, $\mathbf{w}^1$ (i.e., after one iteration of perceptron algorithm) is _____

Solution:-

We are given:

- Class A samples (label +1): $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$
- Class B sample (label -1): $\mathbf{x}_3 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$
- Initial weights: $\mathbf{w}^{(0)} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$
- Learning rate: $\eta = 0.1$

We augment each input vector with a bias term of 1:

$$\tilde{\mathbf{x}}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \tilde{\mathbf{x}}_2 = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}, \quad \tilde{\mathbf{x}}_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

Now we apply one iteration of the Perceptron algorithm:

- For $\tilde{\mathbf{x}}_1, y = +1$:
 $\mathbf{w}^{(0)} \cdot \tilde{\mathbf{x}}_1 = [1, 1, 0] \cdot [1, 2, 1] = 3 > 0 \Rightarrow \text{Correct}$
- For $\tilde{\mathbf{x}}_2, y = +1$:
 $\mathbf{w}^{(0)} \cdot \tilde{\mathbf{x}}_2 = [1, 1, 0] \cdot [4, 2, 1] = 6 > 0 \Rightarrow \text{Correct}$
- For $\tilde{\mathbf{x}}_3, y = -1$:
 $\mathbf{w}^{(0)} \cdot \tilde{\mathbf{x}}_3 = [1, 1, 0] \cdot [3, 0, 1] = 3 > 0 \Rightarrow \text{Incorrect}$

Update rule:

$$\mathbf{w}^{(1)} = \mathbf{w}^{(0)} + \eta \cdot y \cdot \tilde{\mathbf{x}}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + 0.1 \cdot (-1) \cdot \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 1 \\ -0.1 \end{bmatrix}$$

Final answer:

$$\boxed{\mathbf{w}^{(1)} = \begin{bmatrix} 0.7 \\ 1 \\ -0.1 \end{bmatrix}}$$

-
5. Consider a 1D binary classification problem (2.5,+), (4.2,+), (5.6,+), (7.8,-). We train a hard margin SVM. The support vector(s) is/are _____ (if there are more than one SV, write all.)

Solution:-

We are given the one-dimensional dataset:

$$\{(2.5, +), (4.2, +), (5.6, +), (7.8, -)\}.$$

For a hard-margin SVM with decision function

$$f(x) = wx + b,$$

we must satisfy the constraint

$$y_i(wx_i + b) \geq 1 \quad \text{for all } i.$$

Intuition:

- The positive points 2.5, 4.2, and 5.6 lie to the left, while the negative point 7.8 is to the right.
- To maximize the margin while correctly separating the classes, the decision boundary is determined by the closest points to it—namely, the rightmost positive example and the negative example.
- This implies that the point at $x = 5.6$ (with label $+$) and the point at $x = 7.8$ (with label $-$) will be the support vectors, satisfying:

$$w \cdot 5.6 + b = +1 \quad \text{and} \quad w \cdot 7.8 + b = -1.$$

Thus, the support vectors that define the optimal margin are:

$(5.6, +) \quad \text{and} \quad (7.8, -).$

6. Consider the following dataset $\mathcal{D}$ of 5 samples in 2D, $\mathcal{D} = \{[1, 1]^T, [2, 2]^T, [3, 3]^T, [4, 5]^T, [5, 4]^T\}$. The covariance of $\mathcal{D}$ is _____

Solution: Computing the 2×2 Sample Covariance Matrix

Given the data set

$$D = \{(1, 1), (2, 2), (3, 3), (4, 5), (5, 4)\},$$

we will compute the 2×2 sample covariance matrix.

Step 1: Compute the Sample Mean

Let the number of data points be $N = 5$. The sample mean is given by

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i.$$

Writing out the coordinates:

$$\bar{x} = \frac{1}{5} \left[(1, 1) + (2, 2) + (3, 3) + (4, 5) + (5, 4) \right].$$

Compute the sum component-wise:

$$\sum_{i=1}^5 x_i = (1 + 2 + 3 + 4 + 5, 1 + 2 + 3 + 5 + 4) = (15, 15).$$

Thus,

$$\bar{x} = \frac{1}{5}(15, 15) = (3, 3).$$

Step 2: Center Each Data Point

Subtract the mean from each point:

$$(1, 1) - (3, 3) = (-2, -2),$$

$$(2, 2) - (3, 3) = (-1, -1),$$

$$(3, 3) - (3, 3) = (0, 0),$$

$$(4, 5) - (3, 3) = (1, 2),$$

$$(5, 4) - (3, 3) = (2, 1).$$

Step 3: Compute the Sum of the Outer Products

The sample covariance matrix is given by

$$\Sigma = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})(x_i - \bar{x})^T.$$

We compute the outer products one-by-one:

(a) For $x_1 - \bar{x} = (-2, -2)$:

$$(-2, -2)^T (-2, -2) = \begin{pmatrix} (-2)(-2) & (-2)(-2) \\ (-2)(-2) & (-2)(-2) \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}.$$

(b) For $x_2 - \bar{x} = (-1, -1)$:

$$(-1, -1)^T (-1, -1) = \begin{pmatrix} (-1)(-1) & (-1)(-1) \\ (-1)(-1) & (-1)(-1) \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

(c) For $x_3 - \bar{x} = (0, 0)$:

$$(0, 0)^T (0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

(d) For $x_4 - \bar{x} = (1, 2)$:

$$(1, 2)^T (1, 2) = \begin{pmatrix} 1 \cdot 1 & 1 \cdot 2 \\ 2 \cdot 1 & 2 \cdot 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

(e) For $x_5 - \bar{x} = (2, 1)$:

$$(2, 1)^T (2, 1) = \begin{pmatrix} 2 \cdot 2 & 2 \cdot 1 \\ 1 \cdot 2 & 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}.$$

Summing all these outer products:

$$\begin{aligned} S &= \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} + \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 4+1+0+1+4 & 4+1+0+2+2 \\ 4+1+0+2+2 & 4+1+0+4+1 \end{pmatrix} \\ &= \begin{pmatrix} 10 & 9 \\ 9 & 10 \end{pmatrix}. \end{aligned}$$

Step 4: Divide by $N - 1$

Since $N = 5$, we have $N - 1 = 4$. The sample covariance matrix is:

$$\Sigma = \frac{1}{4} \begin{pmatrix} 10 & 9 \\ 9 & 10 \end{pmatrix} = \begin{pmatrix} \frac{10}{4} & \frac{9}{4} \\ \frac{9}{4} & \frac{10}{4} \end{pmatrix} = \begin{pmatrix} 2.5 & 2.25 \\ 2.25 & 2.5 \end{pmatrix}.$$

Final Answer

$$\begin{pmatrix} 2.5 & 2.25 \\ 2.25 & 2.5 \end{pmatrix}$$

7. A dataset $\mathcal{D}$ consists of six feature measurements. $\mathcal{D} = \{1.2, 2.5, 3.8, 0.7, 1.8, 2.0\}$. $\mathcal{D}$ needs to be normalized such that mean is zero and variance is unity. The normalized $\mathcal{D}$ is _____

Solution :- Normalization of Dataset D

Let

$$D = \{1.2, 2.5, 3.8, 0.7, 1.8, 2.0\}.$$

Step 1: Compute the Mean

$$\mu = \frac{1.2 + 2.5 + 3.8 + 0.7 + 1.8 + 2.0}{6} = 2.0.$$

Step 2: Compute the Variance and Standard Deviation

$$\begin{aligned}\sigma^2 &= \frac{1}{6} \sum_{i=1}^6 (x_i - \mu)^2 \\ &= \frac{(1.2 - 2.0)^2 + (2.5 - 2.0)^2 + (3.8 - 2.0)^2 + (0.7 - 2.0)^2 + (1.8 - 2.0)^2 + (2.0 - 2.0)^2}{6} \\ &= \frac{0.64 + 0.25 + 3.24 + 1.69 + 0.04 + 0}{6} \approx 0.9767, \\ \sigma &= \sqrt{\sigma^2} \approx 0.9883.\end{aligned}$$

Step 3: Compute the Normalized Values

For each $x_i \in D$, define

$$z_i = \frac{x_i - \mu}{\sigma}.$$

The computations are summarized in the table below:

x_i	$x_i - \mu$	$z_i = \frac{x_i - \mu}{\sigma}$
1.2	-0.8	-0.81
2.5	+0.5	+0.51
3.8	+1.8	+1.82
0.7	-1.3	-1.32
1.8	-0.2	-0.20
2.0	0.0	0.00

Hence, the normalized dataset is

$$\{-0.81, 0.51, 1.82, -1.32, -0.20, 0.00\}.$$

8. We know for a valid kernel $\kappa(\mathbf{p}, \mathbf{q})$, there exists a $\phi(\mathbf{p})^T \phi(\mathbf{q})$ form where ϕ does an appropriate feature transformation on $\mathbf{p}$ and $\mathbf{q}$. When $\mathbf{p} = [3, 1]^T$; $\mathbf{q} = [2, 3]^T$ and $\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q} + 1)^2$, find the corresponding ϕ and write as $\phi(\mathbf{p})^T \phi(\mathbf{q})$ (such as $[- \text{---} - \text{---} - \text{---} - \text{---}]^T [- \text{---} - \text{---} - \text{---} - \text{---}]$)

Solution :-

We are given the kernel:

$$\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q} + 1)^2$$

This is a polynomial kernel of degree 2 with bias term 1. We aim to find a feature transformation $\phi(\cdot)$ such that:

$$\kappa(\mathbf{p}, \mathbf{q}) = \phi(\mathbf{p})^T \phi(\mathbf{q})$$

Let:

$$\mathbf{p} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

The expansion of the kernel is:

$$(\mathbf{p}^T \mathbf{q} + 1)^2 = (p_1 q_1 + p_2 q_2 + 1)^2$$

We can define the feature map ϕ as:

$$\phi(\mathbf{x}) = \begin{bmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ 1 \end{bmatrix}$$

Thus, we compute:

$$\phi(\mathbf{p}) = \begin{bmatrix} 9 \\ 3\sqrt{2} \\ 1 \\ 3\sqrt{2} \\ \sqrt{2} \\ 1 \end{bmatrix}, \quad \phi(\mathbf{q}) = \begin{bmatrix} 4 \\ 6\sqrt{2} \\ 9 \\ 2\sqrt{2} \\ 3\sqrt{2} \\ 1 \end{bmatrix}$$

Finally, the dot product gives:

$$\phi(\mathbf{p})^T \phi(\mathbf{q}) = (3 \cdot 2 + 1 \cdot 3 + 1)^2 = (6 + 3 + 1)^2 = 10^2 = 100$$

To find the feature transformation ϕ such that $\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q} + 1)^2 = \phi(\mathbf{p})^T \phi(\mathbf{q})$, we expand the kernel expression:

$$(\mathbf{p}^T \mathbf{q} + 1)^2 = (p_1 q_1 + p_2 q_2 + 1)^2 = p_1^2 q_1^2 + p_2^2 q_2^2 + 1 + 2p_1 q_1 p_2 q_2 + 2p_1 q_1 + 2p_2 q_2.$$

This corresponds to the dot product of the transformed feature vectors:

$$\phi(\mathbf{p}) = \begin{bmatrix} p_1^2 \\ p_2^2 \\ \sqrt{2}p_1p_2 \\ \sqrt{2}p_1 \\ \sqrt{2}p_2 \\ 1 \end{bmatrix}, \quad \phi(\mathbf{q}) = \begin{bmatrix} q_1^2 \\ q_2^2 \\ \sqrt{2}q_1q_2 \\ \sqrt{2}q_1 \\ \sqrt{2}q_2 \\ 1 \end{bmatrix}.$$

Thus, the required form is:

$$\begin{bmatrix} p_1^2 & p_2^2 & \sqrt{2}p_1p_2 & \sqrt{2}p_1 & \sqrt{2}p_2 & 1 \end{bmatrix}^T \begin{bmatrix} q_1^2 & q_2^2 & \sqrt{2}q_1q_2 & \sqrt{2}q_1 & \sqrt{2}q_2 & 1 \end{bmatrix}.$$

Quiz II

Name:

Roll Number.....

- Answer precisely in the space given. No overwriting. Make assumptions, if really ambiguous.
- Do rough work in the additional sheet.

Fill in the blank with precise answers:

[8 × 5 = 40]

1. Consider a simple neural network with two inputs, one hidden layer with three neurons and one output. No bias. If the input is $x_1 = 4$ and $x_2 = 2$, output is 18 (fact: All weights are 1.0. Activation functions in the hidden layer is ReLU and activation function in the output is linear.)
2. Continuing the about question, assume the target (t) of the above sample was 25, and the loss was squared error, $\frac{\partial \mathcal{L}}{\partial w_{11}}$ is -56 (w_{11} is the weight from first input (i.e., x_1) to first neuron in the hidden layer.)
3. Consider a single layer perceptron with two input and one output. The weights from first and second inputs are w_1 and w_2 respectively. Bias is w_0 . Also assume a ± 1 logic. The activation at the output is: $\phi(x) = +1$ if $x \geq 0$ and -1 else. If $w_0 = 1, w_1 = -1, w_2 = -1$, then this neuron model is equivalent to: NAND.
4. Consider a binary classification problem (A vs B) with perceptron and bias. Consider two samples from A: $[1, 2]^T, [4, 2]^T$ and one sample from B: $[3, 0]^T$. We initialize $\mathbf{w}^0 = [1, 1, 0]^T$. When learning rate $\eta = 0.1$, $\mathbf{w}^1$ (i.e., after one iteration of perceptron algorithm) is $[0.7, 1, -0.1]^T$.
5. Consider a 1D binary classification problem (2.5,+), (4.8,+), (5.8,-), (7.8,-). We train a hard margin SVM. The support vector(s) is/are (4.8, +) and (5.8, -).
6. Consider the following dataset $\mathcal{D}$ of 5 samples in 2D, $\mathcal{D} = \{[1, 2]^T, [2, 1]^T, [3, 3]^T, [4, 5]^T, [5, 4]^T\}$. The covariance of $\mathcal{D}$ is

$$\begin{pmatrix} 2.5 & 2.0 \\ 2.0 & 2.5 \end{pmatrix}$$

or

$$\begin{pmatrix} 2.0 & 1.6 \\ 1.6 & 2.0 \end{pmatrix}$$

7. A dataset $\mathcal{D}$ consists of six feature measurements. $\mathcal{D} = \{4.2, 5.5, 6.8, 3.7, 4.8, 5.0\}$. $\mathcal{D}$ needs to be normalized such that mean is zero and variance is unity. The normalized $\mathcal{D}$ is

$$\{-0.81, 0.51, 1.82, -1.32, -0.20, 0.00\}.$$

8. We know for a valid kernel $\kappa(\mathbf{p}, \mathbf{q})$, there exists a $\phi(\mathbf{p})^T \phi(\mathbf{q})$ form where ϕ does an appropriate feature transformation on $\mathbf{p}$ and $\mathbf{q}$. When $\mathbf{p} = [3, 3]^T$; $\mathbf{q} = [4, 4]^T$ and $\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q} + 1)^2$, find the corresponding ϕ and write as $\phi(\mathbf{p})^T \phi(\mathbf{q})$ (such as $[- \text{---} - \text{---} - \text{---} - \text{---} - \text{---}]^T [- \text{---} - \text{---} - \text{---} - \text{---} - \text{---}]$)

Answer:

The kernel can be expanded as follows.

$$\begin{aligned}\kappa(\mathbf{p}, \mathbf{q}) &= (\mathbf{p}^T \mathbf{q} + 1)^2 \\ &= (p_1 q_1 + p_2 q_2 + 1)^2 \\ &= p_1^2 q_1^2 + 2p_1 p_2 q_1 q_2 + p_2^2 q_2^2 + 2p_1 q_1 + 2p_2 q_2 + 1\end{aligned}$$

We would like to separate the function $\kappa(\mathbf{p}, \mathbf{q})$ as $\phi(\mathbf{p})^T \phi(\mathbf{q})$. Since the function is symmetric in $\mathbf{p}$ and $\mathbf{q}$, this can be achieved by matching the terms in the sum as follows.

$$\phi(\mathbf{p}) = \begin{bmatrix} p_1^2 \\ \sqrt{2} p_1 p_2 \\ p_2^2 \\ \sqrt{2} p_1 \\ \sqrt{2} p_2 \\ 1 \end{bmatrix}$$

i.e.

$$\phi(\mathbf{p}) = \begin{bmatrix} 9 \\ 9\sqrt{2} \\ 9 \\ 3\sqrt{2} \\ 3\sqrt{2} \\ 1 \end{bmatrix}$$

and

$$\phi(\mathbf{q}) = \begin{bmatrix} q_1^2 \\ \sqrt{2} q_1 q_2 \\ q_2^2 \\ \sqrt{2} q_1 \\ \sqrt{2} q_2 \\ 1 \end{bmatrix}$$

i.e.

$$\phi(\mathbf{q}) = \begin{bmatrix} 16 \\ 16\sqrt{2} \\ 16 \\ 4\sqrt{2} \\ 4\sqrt{2} \\ 1 \end{bmatrix}$$

Note that since inner product is invariant to permutation, any permutation of the feature transformation would meet the requirement.

MidSem Question Paper & Key (Combined)

3 Ques. 90 marks 90min

(quest x marks)

Q1) $10 \times 3 = 30$

Q2) $5 \times 6 = 30$

Q3) $1 \times 30 = 30$

Q1: For the following questions, circle the right answer. (None, One or More is Correct). [30 min; $10 \times 3 = 30$ points]

- Consider X to be a square matrix of size $n \times n$ and $X = UDV^T$. (i.e., SVD)
 - Both $X^T X$ and XX^T have the same eigenvalues
 - Both $X^T X$ and XX^T have the same eigenvectors
 - X , $XX^T X$ and XX^T have the same eigenvalues
 - D^2 contains the eigenvalues of $X^T X$ on its diagonal
 - D contains the eigenvalues of $X^T X$ on its diagonal
- Consider X to be a square matrix of size $n \times n$ and $X = UDV^T$. Then:
 - If $\text{rank}(X) = n$, D has all non-zero entries in diagonal.
 - If $\text{rank}(X) = k$, D has k zeros in diagonal
 - If $\text{rank}(X) = k$, D has $n - k$ zeros in diagonal
 - if $\text{rank}(X) = n$ but $|A|$ is a very small number then, D takes the form $D = \text{diag}(d_1, d_2, \dots, \epsilon)$ where ϵ is a very small number
 - None of these
- Given a set of 2D points X on the vertical line $x_2 = 5$, $X = \{[1, 5]^T, [2, 5]^T, [3, 5]^T, [4, 5]^T, [5, 5]^T\}$
We compute the covariance matrix, and its eigen values and eigen vectors. Then:
 - $\lambda_1 \geq \lambda_2$
 - μ is on the same line.
 - Σ is singular
 - Σ is diagonal
 - None of the above.
- (Use notations and conventions from the class) Consider the problem of linear regression where we minimize the loss

$$\mathcal{L}_1 = \frac{1}{N} \sum_{i=1}^N \alpha_i (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \lambda_1 g(\mathbf{w})$$

where $g()$ is a regularization term. We also write the loss in matrix form as

$$\mathcal{L}_2 = \frac{1}{N} [Y - \mathbf{X}\mathbf{w}]^T A [Y - \mathbf{X}\mathbf{w}] + \lambda_2 g(\mathbf{w}).$$

Now: see $\mathcal{L}_2$ closely,

- When A is a diagonal matrix, this is equivalent to weighing each sample independently as in $\mathcal{L}_1$
- When A is not a diagonal matrix, this loss does not make any sense. Don't ever use.
- When A is PD, we can do Cholesky decomposition of A as LL^T and an equivalent formulation is possible in $\mathcal{L}_1$ is each sample getting transformed as $\mathbf{L}^T \mathbf{x}_i$
- When A is a rank deficient matrix, an equivalent formulation is possible in $\mathcal{L}_1$ with a dimensionality reduction of all the $\mathbf{x}_i$.
- None of the above

5. If $\mathbf{A}$ is a $n \times n$ matrix, with every pair of columns orthogonal i.e., $\mathbf{a}_i \cdot \mathbf{a}_j = \mathbf{0} \quad \forall i, j$ and $\|\mathbf{a}_i\| = 1$. Then:
- (a) $\mathbf{A}^{-1} = \mathbf{A}^T$.
 - (b) $\mathbf{A}\mathbf{A}^T = \mathbf{I}$
 - (c) $\mathbf{A}\mathbf{A}^T$ has only one 1 in every column and all others zero.
 - (d) $\mathbf{A}^{-1}$ has only one 1 in every column and all others zero.
 - (e) none of the above
6. Overfitting can be reduced if
- (a) we can have more training data
 - (b) we can reduce the amount of training data
 - (c) we reduce the dimensionality of all samples (i.e., $\mathbf{x}$)
 - (d) we reduce the learning rate (η) in gradient descent
 - (e) we increase the learning rate (η) in gradient descent.
7. A covariance matrix is always:
- (a) square
 - (b) full rank (i.e., rank = d)
 - (c) triangular
 - (d) PSD
 - (e) PD
 - (f) has at least one eigen value negative
 - (g) no eigen value is imaginary
 - (h) all eigen values are non-negative
8. Let $\mathbf{x}_i \in \mathbb{R}^3$. A set of N points $\mathbf{x}_i$ are on a line. Then rank of a matrix $A = \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^T$ is:
- (a) 1
 - (b) 2
 - (c) 3
 - (d) N
 - (e) ∞
9. Gradient Descent rule (for minimizing a differentiable loss)
- (a) says that the solution needs to be updated along the negative gradient of loss function.
 - (b) says that the solution needs to be updated along the gradient of loss function.
 - (c) can be used for solving non-convex optimization problems (as in neural networks)
 - (d) can be used for solving convex optimization problems (as in linear regression)
 - (e) all of the above.
10. Identify the true statements:
- (a) K Nearest Neighbour is a lazy algorithm for classification
 - (b) All real data is truly multivariate Gaussian
 - (c) L_∞ regression provides the best sparse solution
 - (d) L2 regularization is popularly used because it provides differentiability

Brief Reasons for Each Question

Each correct option is equally weighted. For example, if 2 out of 3 correct options were marked, +2 marks will be awarded for that question.

1. **Question 1:** Correct options: (a), (d)

(a) *Both $X^T X$ and XX^T have the same eigenvalues.*

This follows from the fact that $X^T X$ and XX^T share the same non-zero eigenvalues (they have the same singular values).

(d) *X contains the eigenvalues of $X^T X$ on its diagonal.*

In the SVD $X = UDV^T$, the diagonal entries of D are the singular values, which are the square roots of the eigenvalues of $X^T X$.

2. **Question 2:** Correct options: (a), (d) or (a), (c), (d)

(a) Often states that the rank of X is related to the non-zero singular values.

(c) If included, typically addresses that X has a particular form/behavior (e.g., full column rank) that connects B or D to rank properties.

(d) Can involve diagonalization arguments or how the nullity and rank interplay with X and its factors.

The ambiguity in (d) due to —A— not being defined anywhere is handled by awarding (a),(d) or (a),(c),(d)

3. **Question 3:** Correct options: (a), (b), (c), (d)

(a)–(d) This question concerns a set of 2D points where all points share the same second coordinate $x_2 = 5$, leading to dependencies among the coordinates.

- The covariance matrix is computed as:

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & 0 \\ 0 & 0 \end{bmatrix}$$

where $\text{Var}(X_1)$ is the variance of the first coordinate values $\{1, 2, 3, 4, 5\}$.

- The variance of X_1 is given by:

$$\text{Var}(X_1) = \frac{1}{5} \sum_{i=1}^5 (x_i - \bar{x})^2 = 2.$$

- Thus, the covariance matrix simplifies to:

$$\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

- The eigenvalues of Σ are the diagonal entries:

$$\lambda_1 = 2, \quad \lambda_2 = 0.$$

- Since $\lambda_1 > \lambda_2$, statement (a) holds.
4. **Question 4:** Correct options: (a), (c), (d)
- (a) Increasing training data helps reduce overfitting.
 - (c) Reducing model complexity (e.g. fewer parameters/features) also combats overfitting.
 - (d) Proper regularization can reduce overfitting by penalizing large weights or complex models.
5. **Question 5:** Correct options: (a), (b)
- If A is an $n \times n$ matrix with orthonormal columns, then $A^T A = I$ and $\|a_i\| = 1$.
 - (a) Typically $A^T = A^{-1}$ for an orthonormal (orthogonal) matrix.
 - (b) Each column is unit norm and orthogonal to the others, matching the given conditions.
6. **Question 6:** Correct options: (a), (c), (d)
- Overfitting can be reduced by:
 - (a) Getting more data,
 - (c) Increasing regularization or decreasing model complexity,
 - (d) Early stopping or other gradient-based methods that prevent overfitting.
7. **Question 7:** Correct options: (a), (d), (g), (h)
- As per definitions
8. **Question 8:** Correct option: (a)
- If the set of points $\{x_i\}$ lies on a single line, then the rank of the sum $\sum x_i x_i^T$ is 1. Hence only (a) is correct.
9. **Question 9:** Correct options: (a), (c), (d)
- (a) Gradient descent requires the loss to be differentiable for a direct gradient-based update.
 - (c) It is used for optimizing continuous parameters in many ML problems.
 - (d) It can be extended for non-convex problems, though convergence is not guaranteed to the global optimum.
10. **Question 10:** Correct options: (a), (d)
- (a) k -Nearest Neighbors is indeed a “lazy” learning algorithm.
 - (d) L_2 regularization is a common choice (e.g. ridge regression) that is popular because it is differentiable and helps control model complexity.

Answer to Q2.3: Relationship between the eigenvalues of $\mathbf{A}^T \mathbf{A}$ and the singular values of $\mathbf{A}$

Given: The Singular Value Decomposition (SVD) of a matrix $\mathbf{A}$ is

$$\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{V}^T,$$

where:

- $\mathbf{U}$ is an orthogonal matrix ($\mathbf{U}^T \mathbf{U} = \mathbf{I}$).
- $\mathbf{V}$ is an orthogonal matrix ($\mathbf{V}^T \mathbf{V} = \mathbf{I}$).
- $\mathbf{D}$ is a diagonal matrix whose diagonal entries are the singular values $\sigma_i \geq 0$.

Derivation:

$$\mathbf{A}^T \mathbf{A} = (\mathbf{U} \mathbf{D} \mathbf{V}^T)^T (\mathbf{U} \mathbf{D} \mathbf{V}^T) = \mathbf{V} \mathbf{D}^T \mathbf{U}^T \mathbf{U} \mathbf{D} \mathbf{V}^T = \mathbf{V} \mathbf{D}^T \mathbf{D} \mathbf{V}^T = \mathbf{V} \mathbf{D}^2 \mathbf{V}^T,$$

since $\mathbf{U}^T \mathbf{U} = \mathbf{I}$ and $\mathbf{D}^T = \mathbf{D}$ (diagonal).

- The columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{A}^T \mathbf{A}$.
- The diagonal entries of $\mathbf{D}^2$ (i.e. σ_i^2) are the corresponding eigenvalues of $\mathbf{A}^T \mathbf{A}$.

Hence, if σ_i are the singular values of $\mathbf{A}$, and λ_i are the eigenvalues of $\mathbf{A}^T \mathbf{A}$, then:

$$\lambda_i = \sigma_i^2.$$

Marking Scheme (6 marks)

1. Writing $\mathbf{A}^T \mathbf{A} = (\mathbf{U} \mathbf{D} \mathbf{V}^T)^T (\mathbf{U} \mathbf{D} \mathbf{V}^T)$ correctly 1 mark
2. Simplifying to $\mathbf{A}^T \mathbf{A} = \mathbf{V} \mathbf{D}^T \mathbf{U}^T \mathbf{U} \mathbf{D} \mathbf{V}^T$ 1 mark
3. Recognizing $\mathbf{U}^T \mathbf{U} = \mathbf{I}$ due to orthogonality 1 mark
4. Further simplifying to $\mathbf{A}^T \mathbf{A} = \mathbf{V} \mathbf{D}^T \mathbf{D} \mathbf{V}^T$ 1 mark
5. Recognizing $\mathbf{D}^T = \mathbf{D}$ (since $\mathbf{D}$ is diagonal) and getting $\mathbf{A}^T \mathbf{A} = \mathbf{V} \mathbf{D}^2 \mathbf{V}^T$ 1 mark
6. Stating the final relationship: $\lambda_i = \sigma_i^2$ (the eigenvalues of $\mathbf{A}^T \mathbf{A}$ are the squares of the singular values of $\mathbf{A}$) 1 mark

Note: Partial marks can be awarded for correct steps with minor errors or alternative valid derivations.

Q2. Answer briefly in the space provided[30 min; $5 \times 6 = 30$ points]

1. A test for a certain rare disease is assumed to be correct 95% of the time: if a person has (does not have) the disease, the test results are positive (negative) with probability 0.95. A random person drawn from a certain population has probability 0.001 of having the disease. Given that the person just tested positive, what is the probability that the person has the disease?

Solution:

Given:

$$P(\text{testing positive} \mid \text{person has the disease}) = 0.95$$

$$P(\text{testing negative} \mid \text{person does not have the disease}) = 0.95$$

$$\Rightarrow P(\text{testing positive} \mid \text{person does not have the disease}) = 1 - 0.95 = 0.05$$

$$P(\text{having the disease}) = 0.001$$

$$\Rightarrow P(\text{not having the disease}) = 1 - 0.001 = 0.999$$

Bayes theorem states:

$$P(B|A) = \frac{P(B)P(A|B)}{P(A)}$$

$$P(A) = P(B)P(A|B) + P(\text{not } B)P(A|\text{not } B)$$

In the given question:

 A = testing positive B = having the disease

Substituting in the equation we get:

$$P(\text{having the disease} \mid \text{testing positive}) = \frac{0.001 \times 0.95}{(0.001 \times 0.95) + (0.999 \times 0.05)}$$
$$= \frac{0.00095}{0.0509} \approx 0.0187$$

Marks are deducted for:

- Not writing the Bayesian formula
- Incorrect substitution
- Early approximation of values (0.04995 to 0.05, etc)
- Missing intermediate steps or final answer (0.01866)

2. Consider a two class classification problem with respective means as μ_A and μ_B . Both are univariate Gaussian with variance σ^2 . The optimal threshold/classifier is θ . If prior probability of class B is twice as that of class A, will θ be close to μ_A or μ_B ? (one word answer). Draw a neat simple diagram and demonstrate.

Solution: μ_A

- 2 marks for correct answer
- 1 mark for drawing 2 Gaussians
- 1 mark for same width of both curves (same variance)
- 1 mark for height difference
- 1 mark for showing correct decision boundary

Any of the below diagrams is accepted:

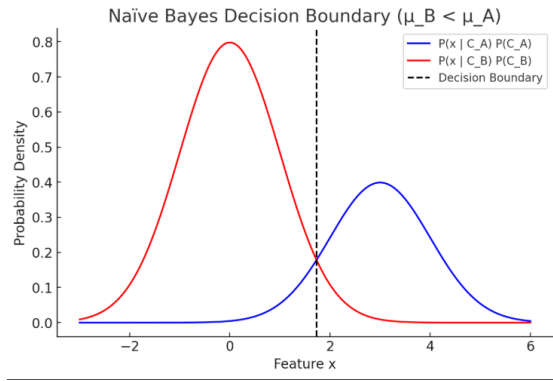


Figure 1

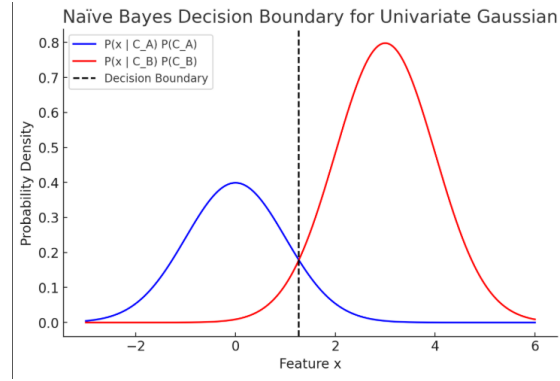


Figure 2

- 3) Consider the SVD of $\mathbf{A}$ as $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T$, Derive a relationship between eigen values of $\mathbf{A}^T\mathbf{A}$ and Singular values of $\mathbf{A}$.

Solution:

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T$$

$$\Rightarrow \mathbf{A}^T\mathbf{A} = (\mathbf{U}\mathbf{D}\mathbf{V}^T)^T(\mathbf{U}\mathbf{D}\mathbf{V}^T)$$

$$\Rightarrow \mathbf{A}^T\mathbf{A} = \mathbf{V}\mathbf{D}^T\mathbf{U}^T\mathbf{U}\mathbf{D}\mathbf{V}^T$$

$$\Rightarrow \mathbf{A}^T\mathbf{A} = \mathbf{V}\mathbf{D}^T\mathbf{D}\mathbf{V}^T \quad [\text{Since } \mathbf{U} \text{ is orthogonal, } \mathbf{U}\mathbf{U}^T = \mathbf{U}^T\mathbf{U} = \mathbf{I}]$$

By properties of SVD decomposition

$\mathbf{D}$ is diagonal matrix of singular values of $\mathbf{A}$

$\Rightarrow \mathbf{D}^T\mathbf{D}$ is diagonal matrix of squared singular values of $\mathbf{A}$

Thus $\mathbf{A}^T\mathbf{A} = \mathbf{V}(\mathbf{D}^T\mathbf{D})\mathbf{V}^T$ forms the eigenvalue decomposition

i.e. $\mathbf{D}^T\mathbf{D}$ is diagonal matrix of eigenvalues of $\mathbf{A}^T\mathbf{A}$

From the above two statements, we get the relation -

The eigenvalues of $\mathbf{A}^T\mathbf{A}$ are the same as the squared singular values of $\mathbf{A}$

- 4) Derive a gradient descent update equation for the minimizing the below loss function

$$\mathcal{L}(\mathbf{w}) = \mathbf{w}^T\mathbf{A}\mathbf{w} - 8(\mathbf{w}^T\mathbf{w} - 1) + 25$$

Solution:

Compute the Gradient of the Loss Function

To derive the gradient compute the partial derivatives of $L(w)$ with respect to w :

1. The gradient of $w^T\mathbf{A}w$ with respect to w is:

$$\bullet_w (w^T\mathbf{A}w) = 2\mathbf{A}w$$

(Note that $\mathbf{A}$ is assumed to be a symmetric matrix).

2. The gradient of $w^T w$ with respect to w is $2w$, so the gradient of this term is:

$$\bullet_w (-8(w^T w - 1)) = -16w$$

3. The gradient of a constant term is zero.

Combining the Gradients,

$$\bullet_w L(w) = 2\mathbf{A}w - 16w$$

Equation to update parameters:

$$w_{\text{new}} = w - \eta \bullet_w L(w)$$

where η is the learning rate.

$$w_{\text{new}} = w - \eta(2Aw - 16w)$$

Simplifying:

$$w_{\text{new}} = w - \eta((2A - 16I)w)$$

where I is the identity matrix of the same dimension as A .

- 1 mark for correct gradient descent formula
- 3 marks for calculating derivative of each term correctly
- 1 mark for correct final update equation
- 1 mark for overall understanding
- (*This question has been marked on a deductive basis*)

5. We know often matrices have low rank. We use this effectively in machine learning

(a) Complete the matrix A if it is of rank-1

$$A = \begin{bmatrix} ?, ?, 2, ?, ? \\ ?, 4, ?, 6, ? \\ ?, 2, ?, ?, 1 \\ 3, ?, ?, ?, ? \\ 1, ?, 1, ?, 1 \end{bmatrix} \Rightarrow$$

(b) Find the rank 1 approximation (nearest rank 1 matrix) of A ¹

$$A = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

Solution:

Note - the marks distribution for this question is 2 and 4 respectively for parts (a) and (b) (a) Matrix of rank 1 implies that all the rows are scalar multiples of each other

Pairwise compare rows and use proportion to compute the missing values of the row

For example, comparing the 3rd and the 5th row -

$$\frac{?}{1} = \frac{2}{?} = \frac{?}{1} = \frac{?}{?} = \frac{1}{1}$$

$$\Rightarrow \text{3rd row} = \text{5th row} = [1 \quad 2 \quad 1 \quad ? \quad 1]$$

Final answer:

$$A = \begin{bmatrix} 2, 4, 2, 6, 2 \\ 2, 4, 2, 6, 2 \\ 1, 2, 1, 3, 1 \\ 3, 6, 3, 9, 3 \\ 1, 2, 1, 3, 1 \end{bmatrix}$$

$$^1\text{SVD of } A = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 4 & 0 & 0 \\ 0 & \sqrt{6} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

(b) Given the SVD decomposition

$$A = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 4 & 0 & 0 \\ 0 & \sqrt{6} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

For the rank 1 approximation, we will take the maximum singular value of A as σ , and its corresponding vectors from U and V of the SVD decomposition as u and v respectively

$\therefore$ rank 1 approximation $A' = \sigma_1 uv^T$

$$= 4 \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= 4 \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 2 \\ 2 & 0 & 2 \end{bmatrix}$$

Q3. Eigen Face Algorithm: Here, question as well as answer is provided!!. You only need to fill in the blanks; You are given many options to choose from at the end ²

If the given options are insufficient (which surely is) or steps are missing or reordering, please add/edit precisely. If there is a step missing, add at the end and edit numbering (eg. add a step 7b, if a step is missing 7 and 8) [30 min; 30 point]

You are given 1000 images of frontal face cropped from 200 individuals (5 distinct faces per person). Each image is of size 200×200 pixel size. We are now given a query face image $\mathbf{a}$ of size 200×200 . We need to compute the most similar 10 faces to $\mathbf{a}$.

Answer Key:

1. Flatten all the images into a vector $\mathbf{x}_i$ of $d = \underline{40K}$. There are a total of $N = \underline{1000}$ samples for computing the dimensionality reduction matrix A for PCA.
2. Compute mean $\mu = \underline{\frac{1}{N} \sum_{i=1}^N x_i}$ from these samples. Subtract mean from all samples to normalize $\mathbf{x}' = \mathbf{x} - \mu$
3. Arrange all the mean subtracted samples as columns and create a data matrix X of size $\underline{d \times N}$.
4. Compute covariance matrix Σ from X as $\underline{\frac{1}{N} X X^T}$. Note that the dimension of Σ is $\underline{d \times d}$.
5. This Σ matrix can have a maximum of $\underline{N = 1000}$ non-zero eigenvalues.
6. Computing eigenvectors of Σ is costly; to simplify the computation, we first find a smaller matrix C of size $\underline{N \times N}$ from X as $C = \underline{X^T X}$.
7. We know that the eigen values of C and Σ are related: they are the same.
8. We compute eigen vectors of C as $\mathbf{u}$ of size $\underline{N \times 1}$, and compute eigen vectors $\mathbf{v}$ of Σ of size $\underline{d \times 1}$. This is done using a simple multiplication as $\mathbf{v} = \underline{X \mathbf{u}}$.
9. We choose k prominent eigen vectors by analyzing the eigen value spectrum. We found $k = 100$
10. We arrange the k eigen vectors as rows and form a matrix A of size $\underline{k \times d}$.
11. New reduced-dimensional vectors: $\mathbf{p}_i = \underline{A \mathbf{x}'_i}$.

² $(\cdot)10(\cdot)100(\cdot)10^4(\cdot)10^6(\cdot)40K(\cdot)N \times N(\cdot)d \times d(\cdot)N \times d(\cdot)d \times N(\cdot)\frac{1}{N} X X^T(\cdot)\frac{1}{N} X^T X(\cdot)\frac{1}{N} \sum_{i=1}^N \mathbf{x}_i(\cdot)$ same $(\cdot)$ scaled versions $(\cdot)k \times N(\cdot)k \times d(\cdot)A \mathbf{x}'_i(\cdot)A \mathbf{x}'_i$

12. We now have N reduced dimensional samples $\mathbf{p}_1, \dots, \mathbf{p}_N$ of size $\underline{k}$.
13. Similarly we compute the reduced dimensional representation of $\mathbf{a}$: $\mathbf{q} = \underline{A(\mathbf{a} - \mu)}$.
14. We compute the distance of $\mathbf{q}$ from all $\mathbf{p}_i$ as $d_i = \underline{\|q - p_i\|^2}$.
15. We sort d_i and find the $\mathbf{p}_i$ corresponding to the smaller 10 distances, and return the corresponding faces.

EndSem Question paper & Key

3hrs , 3 Ques, 180 marks

$$Q1) 12 \times 5 = 60$$

$$Q2) 3 \times 20 = 60$$

$$Q3) 3 \times 20 = 60$$

Part I: Simple objective questions; No details/steps needed [60 min; $12 \times 5 = 60$ points]

1. If a neural network designed for classification outputs $[0.2, 0.1, 0.7, 0.0]^T$ and the target is $[0, 0, 1, 0]^T$. Compute the cross entropy loss.

Expression: _____

$$- y_i \cdot \log(\hat{y}_i)$$

Numerical Value: _____

$$\log(0.7)$$

2. Consider a 1D convolution. A convolution layer has 3 input channel of size 1000 each. Output is computed over a 1-D window of length 7. There are 5 output channels. Stride is 3. No bias.

How many learnable parameters exists in this layer?

Answer: $7 \times 3 \times 5$ or 110 for bias

3. Make the necessary minimal changes (if any required) and rewrite as true sentence in the space provided. Avoid changing the words in bold.

Adding a momentum term *removes the problem of local minima* **in Neural network training.**

Adding a momentum term reduces the problem of local minima in Neural network training.

4. Consider the following question with multiple answers. Given answers could be incorrect. Correctly answer in the space given with an answer approximately of equal length/detail, preferably by rewording or copying one of the given answers. Pick the closest answer and consider rewriting.

Q: Given the objective/loss of a neural network, why don't we differentiate with respect to the individual weights, equate to zero and find the optimal (may be local) solution in one step?

Ans1: Yes; it is feasible. We do not do this because back-propagation is simpler to implement in python.

Ans2: No; this is not possible. Because we do not know how to differentiate the loss with respect to individual weights.

Possible but not feasible in reality because solving a non linear equation with a large number of variables (number of weights) is almost impossible

Your Answer: _____

Ans 2.(1 mark) correct explanation(4 mark)

5. Consider the following question with multiple answers. Given answers could be incorrect. Correctly answer in the space given with an answer approximately of equal length/detail, preferably by rewording or copying one of the given answers. Pick the closest answer and consider rewriting.

Q: We can initialize all the weights of an MLP as equal (say 1.0). This is considered to be bad. Why?

Ans1: No; This is a perfectly fine initialization

Ans2: Yes; this is not a good initialization. Since weights are equal, we can not find the derivatives

Ans3: Yes; this is not a good initialization. Since weights are equal, all weights learn at constant speed and we will not get a sparse solution as network.

Not a good initialization, all nodes of a layer for all layers except first layer will have same derivative and thus weights will remain always same across a layer. "Symmetry" needs to be broken.

6. Consider the following question with multiple answers. Given answers could be incorrect. Correctly answer in the space given with an answer approximately of equal length/detail, preferably by rewording or copying one of the given answers. Pick the closest answer and consider rewriting.

Q: Consider an autoencoder where input and output are x . We train this network by minimizing the MSE loss of predicted and actual, using BP. Why does not loss become zero at the end of the training?

Ans1: Neural networks can not learn identity function.

Ans2: Training by backpropagation will never allow to make loss zero or near zero.

Ans3: The reason is the architecture. With more layers (deeper network), the loss will become zero or near zero.

Loss does not become zero because we only want to minimize the loss on a non-convex surface locally in a neural network. [OR]
bottleneck layer inability to capture data perfectly causes non-zero loss”

Your Answer: _____

7. Consider the following code snippet for convolution

```
conv = nn.Conv1d(in_channels=1,out_channels=5,
                 kernel_size=7,padding=3,bias=True)
input = torch.randn(12,1,100)
output = conv(input)
```

What is the number of trainable parameters in conv? What is the “output” dimension?

Ans: 40, 12 * 5 * 100

8. Consider this model:

```
class Net(nn.Module):
    def __init__(self):
        super(Net, self).__init__()
        self.fc1 = nn.Linear(10,4, bias=False)
        self.fc2 = nn.Linear(4, 2, bias=True)
    def forward(self, x):
        x = self.fc1(x)
        x = nn.ReLU()(x)
        x = self.fc2(x)
        return x
```

What is the number of trainable parameters in this model

Ans: _____

50

9. Consider a problem of deciding an email as SPAM or NON-SPAM. We have a trained classifier/solution that scores each email as $\mathbf{w}^T \mathbf{x}$. However, our job will be to define a threshold so that we can say $\mathbf{w}^T \mathbf{x} \geq \theta$ for SPAMs

For the 10 emails that we have the scores (i.e., $\mathbf{w}^T \mathbf{x}$) are

0.7, 0.65, 0.92, 0.75, 0.45, 0.8, 0.3, 0.8, 0.1, 0.1

and we know the truth/label as (SPAM (+) and NON-SPAM (-))

+ , - , + , + , - , + , + , + , - , -

For θ as 0.5 and 0.2, find the “Precision” and “Recall”.

Ans: (0.5): Precision: _____; Recall: _____

Ans: (0.2): Precision: _____; Recall: _____

10. Identify, none, one or more correct answers:

We define/know the weighted Euclidean distance

$$\tau(\mathbf{x}, \mathbf{y}) = [\mathbf{x} - \mathbf{y}]^T [\mathbf{A}] [\mathbf{x} - \mathbf{y}]$$

Where $\mathbf{x}$ and $\mathbf{y}$ are vectors in d dimension $\mathbf{A}$ is a square matrix

- (a) If $\mathbf{A}$ is a non-identity diagonal matrix. Then “dist” is a scaled version of Euclidean distance or “ $\tau = \alpha$ Euclidean distance”
- (b) If $k < d$ and $A = \sum_{i=1}^k \mathbf{u}_i \mathbf{u}_i^T$, with $\mathbf{u}_i$ s as orthonormal. Then A is rank deficient and Euclidean τ can not be computed.
- (c) If $k < d$ and $A = \sum_{i=1}^k \mathbf{u}_i \mathbf{u}_i^T$, with $\mathbf{u}_i$ s as orthonormal. Then τ is equivalent to dimensionality reduction $\mathbf{x}' = \mathbf{W}\mathbf{x}$ with $\mathbf{W}$ as $k \times d$ matrix with $\mathbf{u}_i^T$ as the i th row.
- (d) When $\mathbf{A}$ is non-diagonal symmetric matrix, $\tau(\mathbf{x}, \mathbf{y}) = \tau(\mathbf{y}, \mathbf{x})$.
- (e) All the above are TRUE.

Ans: ACD

11. Bag I contain w_1 white and b_1 black balls. Bag II contains w_2 white and b_2 black balls.

A ball is drawn at random from one of the bags, and it is found to be white. What is the probability that it was drawn from Bag I.

- (a) $\frac{w_1}{w_1 + b_1}$
- (b) $\frac{w_1(w_1 + b_1)}{w_1(w_1 + b_1) + w_2(w_2 + b_2)}$
- (c) $\frac{w_1(b_1 + b_2)}{w_1(b_1 + b_2) + w_2(b_1 + b_2)}$
- (d) $\frac{w_1(w_2 + b_2)}{w_1(w_2 + b_2) + w_2(w_1 + b_1)}$
- (e) None of the above

Ans: D

12. What is “Self-Supervised Learning” ? Give an example. (not more than 3-4 sentences)

Ans:

Part II: Answer three out of four the following questions [60 min; $3 \times 20 = 60$ points]

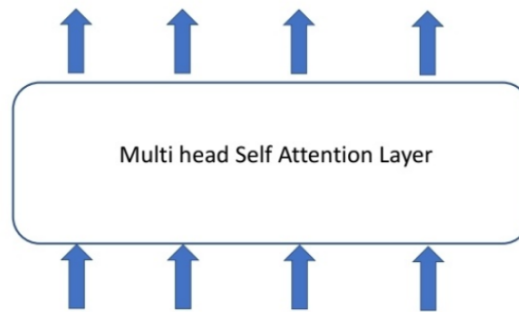
1. **MLP** Consider an MLP neural network with four inputs; one hidden layer of 5 neurons and one output neuron. Hidden layer has ReLU activation and output neuron has linear activation. All neurons in the hidden and output layers have bias.

Let W_1 be the weights from input to hidden layer and W_2 be the weights from hidden to the output layer.

In W_1 , all elements in the odd numbered rows is +1. All elements in the even numbered rows is -1.

In W_2 , all elements are +1. [4 + 4 + 2 + 8 + 2]

- How many trainable weights are there in this neural network?
- Find output for an input $[1, 2, 1, 4]^T$
- If the target (truth) was 15, find a squared error loss $\mathcal{L}$ (i.e., $(t - o)^2$)
- Find $\frac{\partial \mathcal{L}}{\partial v}$, where v is the weight from first input to first hidden neuron.
- Assuming a learning rate of 0.1, write the update equation for v and find the new value numerically.



2. Consider a multi head (two heads) self attention layer. $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4$ are the four inputs and $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3, \mathbf{y}_4$ are the outputs. $\mathbf{x}_i \in \mathbb{R}^4$ (for the simplicity of numerical computation, assume no softmax within the self attention layer)

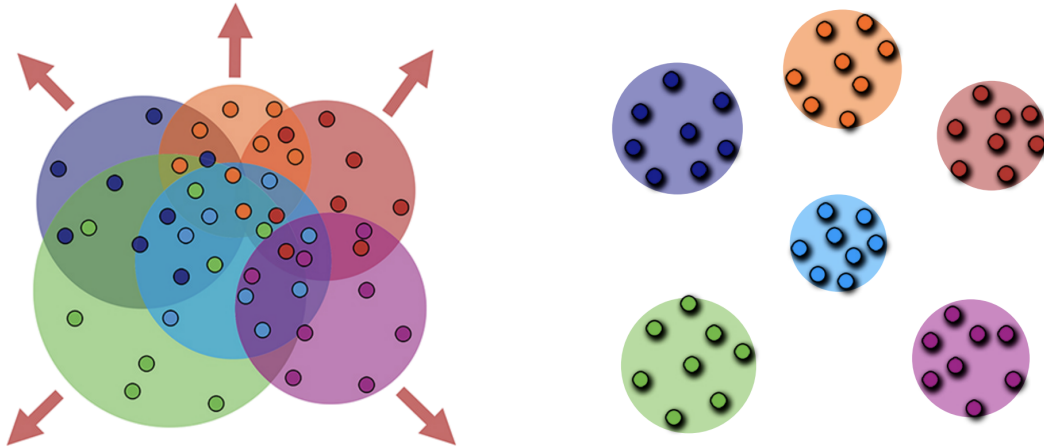
- Head 1: All the three matrices W_Q, W_K and W_V are initialized as identity (square) matrices.
- Head 2: All the three matrices W_Q, W_K and W_V have exactly the same size as of head 1. But W_Q has all its elements 1. Similarly W_K has all its elements -1 and W_V is identity.
- W_O has all its elements initialized as 1. [5 + 8 + 7]

- What are the dimensionality of all the seven learnable matrices. (no bias anywhere)
- Compute $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3, \mathbf{y}_4$ if $\mathbf{x}_1 = [1, 1]^T, \mathbf{x}_2 = [-1, -1]^T, \mathbf{x}_3 = [2, 2]^T, \mathbf{x}_4 = [0, 0]^T$
- Consider a similar simplified architecture to what is shown in figure. but with only two inputs and two outputs; and only one head. no $\mathbf{W}_O$ and W_Q, W_K and W_V are square matrices. Using chain rule find $\frac{\partial \mathcal{L}}{\partial \mathbf{x}_1}$ if we know $\frac{\partial \mathcal{L}}{\partial \mathbf{y}_i}$ (Advice: This question may be more involved; partial grades will be given based on how far you go. Avoid spending too much time.)

3. **Algorithm** You are given N examples $(\mathbf{x}_i, y_i)$. Write the **pseudocode** for stochastic minibatch gradient descent based minimization of the loss $L = \sum_{i=1}^N (y_i - \mathbf{w}^T \mathbf{x}_i)^2$.

Write one word/phrase answer: (i) what are we (which variable) trying to find out? (ii) how do we initialize the iterations? (iii) is this problem convex? (iv) does the final answer depends on initialization/learning rate? (v) Suggest some steps to make sure that the implementation converge and reliable. (vi) suggest an alternate termination criteria (than the one used in the pseudo code).

[4 + 2 + 2 + 2 + 2 + 4 + 4]



4. Design of a loss function: Consider a six class classification problem. We can design a neural network and train them using backpropagation. However, along with minimizing the loss on the given samples, it is perceived as if samples/classes can be well separated as in figure (on right), it is preferred. [5 + 5 + 5 + 5]

- Suggest a loss function for classification of six classes in a neural network setting.
- Suggest how the geometric intuition here (figure) could be expressed analytically as a term to be added to the above loss function.
- Is your loss function differentiable? Can this be computed in a single sample mode?
- We often use a mini-batch SGD for training. Can we have a better mini batch construction (or sampling) for this type of losses?

Design of a Loss Function for Six-Class Classification

(a) Loss function for six-class classification

The most general and widely used loss for multi-class classification is the **Cross-Entropy Loss**. Assuming we use a softmax layer at the output:

Let:

- $\mathbf{z} \in \mathbb{R}^6$ be the logits output from the neural network.
- $\hat{\mathbf{y}} = \text{softmax}(\mathbf{z})$, where

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^6 e^{z_j}}$$

- $y \in \{1, 2, \dots, 6\}$ is the true class label.

Then the standard **Cross-Entropy Loss** is:

$$\mathcal{L}_{\text{CE}} = -\log(\hat{y}_y)$$

This encourages the network to assign high probability to the correct class.

(b) Adding geometric intuition as a regularization term

If the figure suggests geometrically well-separated classes (e.g., clusters or class centers far from each other), we can add a separation-promoting regularization term.

One widely accepted method is to introduce a **Center Loss** or a **Margin-based Loss**. A general form of such a geometric regularizer is:

$$\mathcal{L}_{\text{geom}} = \frac{1}{2} \|\mathbf{f} - \mathbf{c}_y\|^2$$

Where:

- $\mathbf{f} \in \mathbb{R}^d$ is the feature embedding of the sample before the final layer.
- $\mathbf{c}_y$ is the learned center for class y in the feature space.

This term pulls samples of the same class closer to their class center, improving intra-class compactness.

To encourage inter-class separation, we can use:

$$\mathcal{L}_{\text{inter}} = \sum_{i \neq j} \frac{1}{\|\mathbf{c}_i - \mathbf{c}_j\|^2}$$

This penalizes class centers that are too close, encouraging them to spread out in the embedding space.

The total loss becomes:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_1 \mathcal{L}_{\text{geom}} + \lambda_2 \mathcal{L}_{\text{inter}}$$

where λ_1, λ_2 are tunable hyperparameters.

(c) Differentiability and single-sample computation

- **Differentiability:** Yes, the total loss is differentiable. All components — cross-entropy, Euclidean distance, and inverse norm — are differentiable with respect to the network parameters.
- **Single-sample mode:**
 - $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{geom}}$ can be computed per sample.
 - $\mathcal{L}_{\text{inter}}$ requires information about multiple class centers and is typically updated across a batch or over several iterations.

(d) Better mini-batch construction

To effectively train with such a loss, especially those involving both intra-class compactness and inter-class separation, we can use better sampling strategies:

Class-uniform mini-batches: For a dataset with $C = 6$ classes, sample K instances from each class per batch. This ensures:

- Each class center gets sufficiently updated.
- Inter-class terms are computed more reliably.
- Intra-class variation is represented in every batch.

Such sampling is commonly used in metric learning, contrastive learning, and face recognition tasks where geometric losses are crucial.

Part III: Answer three out of four the following questions
points]

[60 min; $3 \times 20 = 60$

1. Given N training samples $\{(\mathbf{x}_i, y_i)\}$ where $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \{-1, +1\}$, geometrically derive SVM (primal) objective. State the constraints.

Given: Training samples:

$$\{(x_i, y_i)\}_{i=1}^N, \quad x_i \in \mathbb{R}^d, \quad y_i \in \{-1, +1\}$$

Step 1: Define the hyperplane

A separating hyperplane is defined as:

$$w \cdot x + b = 0$$

where $w \in \mathbb{R}^d$ and $b \in \mathbb{R}$.

Step 2: Decision function

The decision function for classification:

$$f(x) = \text{sign}(w \cdot x + b)$$

Step 3: Margin calculation

For a point x_i , the distance to the hyperplane is:

$$d(x_i) = \frac{|w \cdot x_i + b|}{\|w\|}$$

For correctly classified points:

$$y_i(w \cdot x_i + b) > 0$$

Step 4: Support vectors and canonical hyperplane

We scale w and b so that for the closest points (support vectors):

$$w \cdot x + b = +1 \quad (\text{positive support vector})$$

$$w \cdot x + b = -1 \quad (\text{negative support vector})$$

Step 5: Margin width calculation

The distance from each margin to the hyperplane is:

$$\frac{1}{\|w\|}$$

So, total margin width is:

$$\frac{2}{\|w\|}$$

Step 6: Classification constraints

For all samples:

$$y_i(w \cdot x_i + b) \geq 1, \quad \forall i \in \{1, 2, \dots, N\}$$

Step 7: Optimization objective

To maximize margin, minimize $\|w\|$, or for convenience:

$$\min_{w, b} \frac{1}{2} \|w\|^2$$

Step 8: Primal SVM (Hard Margin)

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2} \|w\|^2 \\ \text{subject to} \quad & y_i(w \cdot x_i + b) \geq 1, \quad \forall i \end{aligned}$$

Step 9: Soft Margin SVM (Non-separable data)

Introduce slack variables $\xi_i \geq 0$ and penalty parameter C :

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i \\ \text{subject to} \quad & y_i(w \cdot x_i + b) \geq 1 - \xi_i, \quad \forall i \\ & \xi_i \geq 0, \quad \forall i \end{aligned}$$

2. We know that a kernel $\kappa(\mathbf{p}, \mathbf{q}) = \phi(\mathbf{p})^T \phi(\mathbf{q})$.

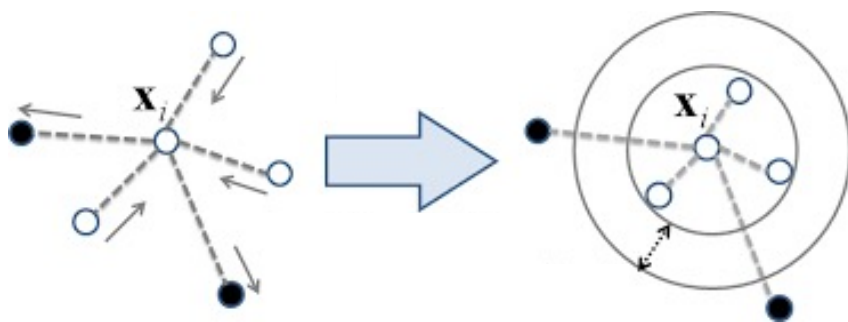
Consider a data set of N samples $\{\mathbf{x}_i\}$ and a Kernel Matrix K such that (i, j) th element of this matrix is $\kappa(\mathbf{x}_i, \mathbf{x}_j)$. [2 + 8 + 10]

- (a) Show that K symmetric
- (b) Show that Kernel matrix (i.e., K) is PSD (Positive Semi Definite)
- (c) Find $\phi(\cdot)$ if $\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q})^3$ and $\mathbf{p} = [p_1, p_2]^T$; $\mathbf{q} = [q_1, q_2]^T$

4. Gradient Descent Optimization

[9 + 2 + 9]

- (a) Consider a function $f(w) = w^2 + 2w + 1$. We want to minimize $f(w)$ using GD. Initialize w^0 as 5. Choose the learning rate η as 0.1. Show two iterations i.e., compute the value of w after first and second iterations.
- (b) Suggest a learning rate better than 0.1 so that it converges faster.
- (c) Suggest a learning rate that will take you to the optimal solution in one step i.e., w^1 . How do we arrive at this “optimal” learning rate, in general?



3. Consider a data set of N samples $\{\mathbf{x}_i\}$ where samples are either positive or negative (two class). (Classes need not be well separated as in the figure). We are using a nearest neighbour algorithm for classification using a simple distance function $d(\mathbf{x}_i, \mathbf{x}_j) = [\mathbf{x}_i - \mathbf{x}_j]^T [\mathbf{x}_i - \mathbf{x}_j]$ [5 + 5 + 5 + 5]
- However, if we can “preprocess” the data as shown in the figure such that all class A (empty circle in figure) come closer, it will help. Let us do this using a feature transformation $\mathbf{z}_i = \mathbf{L}\mathbf{x}_i$. Now, write the distance by including $\mathbf{L}$. Considering just class A: How do we formulate the (maximization/minimization) problem for finding $\mathbf{L}$. (No need to solve. Make sure your function is scalar valued. Check dimensionalities!).
 - What are the pitfalls if $\mathbf{L}$ is NULL (all elements zero)? How do we constrain $\mathbf{L}$ to avoid this pitfall?
 - It is argued that $d(\mathbf{x}_i, \mathbf{x}_j) = [\mathbf{x}_i - \mathbf{x}_j]^T \mathbf{A} [\mathbf{x}_i - \mathbf{x}_j]$ can do this job. If so, relate $\mathbf{A}$ and $\mathbf{L}$
 - To minimize any potential confusion, it is good to give a safe “margin” for all samples in class B (filled circles in figure) from that of class A. Write an objective function and optimization problem to achieve that.

Key for Q2,3

1 Section-1

2 Section-2

2.1 Question-1

2.2 Question-2

Problem Recap:

Consider a two-head self-attention layer with inputs $x_1, x_2, x_3, x_4 \in \mathbb{R}^2$ and outputs $y_1, y_2, y_3, y_4 \in \mathbb{R}^2$. There is no softmax inside the attention.

- **Head 1:** $W_Q^1, W_K^1, W_V^1 = I_{2 \times 2}$.
- **Head 2:** $W_Q^2, W_K^2, W_V^2 = -J_{2 \times 2}$, where J has all entries 1.
- **Output projection:** $W_O \in \mathbb{R}^{2 \times 4}$ with all entries 1.

You were required to answer the following:

1. **Dimensionalities.** What are the sizes of

$$W_Q^1, W_K^1, W_V^1, W_Q^2, W_K^2, W_V^2, W_O?$$

2. **Numerical outputs.** Compute

$$y_i = W_O \left[\underbrace{y_i^{(1)}}_{\text{head1}} \parallel \underbrace{y_i^{(2)}}_{\text{head2}} \right] \quad \text{for } i = 1, 2, 3, 4,$$

where for each head h ,

$$y_i^{(h)} = \sum_{j=1}^4 (Q_i^{(h)} \cdot K_j^{(h)}) V_j^{(h)}, \quad Q_i^{(h)} = W_Q^h x_i, \quad K_j^{(h)} = W_K^h x_j, \quad V_j^{(h)} = W_V^h x_j.$$

3. **Gradients.** In a simplified single-head, two-input analogue (no W_O , all square), use the chain rule to write

$$\frac{\partial \mathcal{L}}{\partial x_1} \quad \text{in terms of} \quad \frac{\partial \mathcal{L}}{\partial y_1}, W_Q, W_K, W_V, \{x_j\}.$$

Solution.

1. Since $x_i \in \mathbb{R}^2$ and each head does

$$Q, K, V : \mathbb{R}^2 \rightarrow \mathbb{R}^2,$$

each of

$$W_Q^1, W_K^1, W_V^1, W_Q^2, W_K^2, W_V^2$$

is 2×2 . We then concatenate two 2-vectors into a 4-vector and apply

$$W_O : \mathbb{R}^4 \rightarrow \mathbb{R}^2,$$

so

$$W_O \in \mathbb{R}^{2 \times 4}.$$

2. **Head 1:** since $W_Q^1 = W_K^1 = W_V^1 = I$,

$$y_i^{(1)} = \sum_{j=1}^4 (x_i \cdot x_j) x_j.$$

A quick table of dot-products and sums gives

$$y_1^{(1)} = [12, 12], y_2^{(1)} = [-12, -12], y_3^{(1)} = [24, 24], y_4^{(1)} = [0, 0].$$

Head 2: here

$$Q_i^{(2)} = Jx_i = (\mathbf{1}^\top x_i) [1, 1]^\top, \quad K_j^{(2)} = -Jx_j = -(\mathbf{1}^\top x_j) [1, 1]^\top,$$

so

$$Q_i^{(2)} \cdot K_j^{(2)} = -2 (\mathbf{1}^\top x_i) (\mathbf{1}^\top x_j).$$

Writing $s_j = \mathbf{1}^\top x_j$, one shows

$$y_i^{(2)} = \sum_{j=1}^4 (-2 s_i s_j) x_j = -2 s_i \sum_{j=1}^4 s_j x_j = \begin{cases} [-48, -48], & i = 1, \\ [48, 48], & i = 2, \\ [-96, -96], & i = 3, \\ [0, 0], & i = 4. \end{cases}$$

Concat & output: for each i , form $[y_i^{(1)} \| y_i^{(2)}] \in R^4$ and multiply by W_O (all ones). Summing the four entries gives

$$y_1 = [-72, -72], y_2 = [72, 72], y_3 = [-144, -144], y_4 = [0, 0].$$

3. **Single-head toy:** let

$$y_i = \sum_{j=1}^2 (W_Q x_i \cdot W_K x_j) (W_V x_j).$$

Then by the chain rule

$$\frac{\partial \mathcal{L}}{\partial x_1} = \sum_{h=Q,K,V} \frac{\partial \mathcal{L}}{\partial y_1} \frac{\partial y_1}{\partial h} \frac{\partial h}{\partial x_1},$$

and one shows for example

$$\frac{\partial y_1}{\partial x_1} = \sum_j \left[(W_V x_j) (W_K x_j)^\top W_Q + (W_Q x_1 \cdot W_K x_j) W_V \right].$$

Thus

$$\frac{\partial \mathcal{L}}{\partial x_1} = \left[\sum_j (W_V x_j) (W_K x_j)^\top W_Q + \sum_j (W_Q x_1 \cdot W_K x_j) W_V \right]^\top \frac{\partial \mathcal{L}}{\partial y_1}.$$

Marking Rubric for Question 2 (20 marks)

2.3 Question-3

You are given N examples $(\mathbf{x}_i, y_i)$. Write the pseudocode for stochastic minibatch gradient descent based minimization of the loss $L = \sum_{i=1}^N (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$.

Part	Criterion	Marks
(a)	Correct sizes for all six head matrices (2×2)	3
	Correct size for W_O (2×4)	2
(b)	Head 1: correct dot-product sums and $y_i^{(1)}$ values	4
	Head 2: correct form of Q, K and numerical $y_i^{(2)}$	4
(c)	Chain-rule structure, identifying all three Jacobian terms	4
	Correct final expression in terms of $W_Q, W_K, W_V, \frac{\partial \mathcal{L}}{\partial y_1}$	3

Pseudocode for Stochastic Minibatch Gradient Descent

Inputs:

- Dataset $D = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)\}$
- Learning rate $\eta > 0$
- Batch size m (where $1 \leq m \leq N$)
- Number of epochs E (or other termination criteria)

Output:

- Optimized weight vector $\mathbf{w}$

```

1: Initialize weight vector  $\mathbf{w}$  (e.g., with small random numbers or zeros).
2: for epoch from 1 to  $E$  do
3:   Shuffle the dataset  $D$ .
4:   for  $j = 0, 1, \dots, \lfloor (N-1)/m \rfloor$  do
5:     Get minibatch  $B_j = \{(\mathbf{x}_k, y_k)\}$  for  $k$  from  $j \cdot m$  to  $\min((j+1) \cdot m - 1, N-1)$ .
6:     Let  $m_j = |B_j|$  be the actual size of the current minibatch.
7:     Calculate the gradient of the loss for the current minibatch  $B_j$ :
8:      $\bullet_{\mathbf{w}} L_{B_j} = \frac{1}{m_j} \sum_{(\mathbf{x}_k, y_k) \in B_j} \frac{\partial (y_k - \mathbf{w}^T \mathbf{x}_k)^2}{\partial \mathbf{w}}$ 
9:      $\bullet_{\mathbf{w}} L_{B_j} = \frac{1}{m_j} \sum_{(\mathbf{x}_k, y_k) \in B_j} 2(y_k - \mathbf{w}^T \mathbf{x}_k)(-\mathbf{x}_k)$ 
10:     $\bullet_{\mathbf{w}} L_{B_j} = -\frac{2}{m_j} \sum_{(\mathbf{x}_k, y_k) \in B_j} (y_k - \mathbf{w}^T \mathbf{x}_k) \mathbf{x}_k$ 
11:    Update the weights:
12:     $\mathbf{w} \leftarrow \mathbf{w} - \eta \bullet_{\mathbf{w}} L_{B_j}$ 
13: return  $\mathbf{w}$ 

```

Write one word/phrase answer

1. **What are we (which variable) trying to find out?**

The weight vector $\mathbf{w}$ (or "optimal weights").

2. **How do we initialize the iterations?**

Initialize $\mathbf{w}$ randomly or to zeros.

3. **Is this problem convex?**

Yes (the loss function is convex).

4. **Does the final answer depends on initialization/learning rate?**

- **Initialization:** No (for convex problems with a unique minimum, it converges to the same global minimum, assuming appropriate learning rate).

- **Learning rate:** Yes (affects convergence speed, stability, and how close to the true optimum the solution gets in a finite number of steps. A very poor learning rate can lead to divergence or a sub-optimal solution).

5. **Suggest some steps to make sure that the implementation converge and reliable.**

- Monitor training/validation loss.
- Tune the learning rate (e.g., use a learning rate schedule, try different values).
- Normalize/standardize input features.
- Gradient checking (to ensure correct gradient implementation).
- Ensure proper shuffling of data each epoch.

6. **Suggest an alternate termination criteria (than the one used in the pseudo code).**

- Stop when the change in loss between epochs falls below a small threshold.
- Stop when the validation loss starts to increase or plateaus (early stopping).
- Stop when the magnitude of the gradient (or change in weights) falls below a small threshold.

2.4 Question-4

3 Section-3

3.1 Question-1

3.2 Question-2

Solution to Question 2

We are given that a kernel $\kappa(\mathbf{p}, \mathbf{q}) = \phi(\mathbf{p})^T \phi(\mathbf{q})$. Consider a data set of N samples $\{\mathbf{x}_i\}$ and a Kernel Matrix K such that the (i, j) -th element of this matrix is $K_{ij} = \kappa(\mathbf{x}_i, \mathbf{x}_j)$.

(a) Show that K is symmetric

A matrix K is symmetric if $K = K^T$, which means $K_{ij} = K_{ji}$ for all i, j . The (i, j) -th element of K is given by:

$$K_{ij} = \kappa(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$$

The (j, i) -th element of K is given by:

$$K_{ji} = \kappa(\mathbf{x}_j, \mathbf{x}_i) = \phi(\mathbf{x}_j)^T \phi(\mathbf{x}_i)$$

Since $\phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$ is a scalar (it's a dot product), its transpose is itself:

$$\phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) = (\phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j))^T$$

Using the property $(AB)^T = B^T A^T$, and noting that $(\phi(\mathbf{x}_i)^T)^T = \phi(\mathbf{x}_i)$:

$$\phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) = \phi(\mathbf{x}_j)^T (\phi(\mathbf{x}_i)^T)^T = \phi(\mathbf{x}_j)^T \phi(\mathbf{x}_i)$$

Thus, we have:

$$K_{ij} = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) = \phi(\mathbf{x}_j)^T \phi(\mathbf{x}_i) = K_{ji}$$

Therefore, K is symmetric.

(b) Show that Kernel matrix (i.e., K) is PSD (Positive Semi Definite)

A symmetric matrix K is Positive Semi-Definite (PSD) if for any non-zero vector $\mathbf{z} \in \mathbb{R}^N$, $\mathbf{z}^T K \mathbf{z} \geq 0$. Let $\mathbf{z} = [z_1, z_2, \dots, z_N]^T$ be an arbitrary non-zero vector.

$$\begin{aligned} \mathbf{z}^T K \mathbf{z} &= \sum_{i=1}^N \sum_{j=1}^N z_i K_{ij} z_j \\ &= \sum_{i=1}^N \sum_{j=1}^N z_i \left(\phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) \right) z_j \\ &= \sum_{i=1}^N \sum_{j=1}^N (z_i \phi(\mathbf{x}_i)^T) (\phi(\mathbf{x}_j) z_j) \\ &= \left(\sum_{i=1}^N z_i \phi(\mathbf{x}_i)^T \right) \left(\sum_{j=1}^N z_j \phi(\mathbf{x}_j) \right) \end{aligned}$$

Let $\mathbf{v} = \sum_{k=1}^N z_k \phi(\mathbf{x}_k)$. Then $\mathbf{v}$ is a vector (in the feature space). The expression above can be written as:

$$\mathbf{z}^T K \mathbf{z} = \mathbf{v}^T \mathbf{v} = \|\mathbf{v}\|^2$$

Since the squared norm of any vector is non-negative, i.e., $\|\mathbf{v}\|^2 \geq 0$, we have:

$$\mathbf{z}^T K \mathbf{z} \geq 0$$

Therefore, the Kernel matrix K is Positive Semi-Definite.

(c) Find $\phi(\cdot)$ if $\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q})^3$ and $\mathbf{p} = [p_1, p_2]^T$; $\mathbf{q} = [q_1, q_2]^T$

We are given $\mathbf{p} = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$. First, calculate $\mathbf{p}^T \mathbf{q}$:

$$\mathbf{p}^T \mathbf{q} = p_1 q_1 + p_2 q_2$$

Now, we need to compute $(\mathbf{p}^T \mathbf{q})^3$:

$$\begin{aligned} \kappa(\mathbf{p}, \mathbf{q}) &= (p_1 q_1 + p_2 q_2)^3 \\ &= (p_1 q_1)^3 + 3(p_1 q_1)^2 (p_2 q_2) + 3(p_1 q_1) (p_2 q_2)^2 + (p_2 q_2)^3 \\ &= p_1^3 q_1^3 + 3p_1^2 p_2 q_1^2 q_2 + 3p_1 p_2^2 q_1 q_2^2 + p_2^3 q_2^3 \\ &= (p_1^3)(q_1^3) + (\sqrt{3}p_1^2 p_2)(\sqrt{3}q_1^2 q_2) + (\sqrt{3}p_1 p_2^2)(\sqrt{3}q_1 q_2^2) + (p_2^3)(q_2^3) \end{aligned}$$

We want to find $\phi(\cdot)$ such that $\kappa(\mathbf{p}, \mathbf{q}) = \phi(\mathbf{p})^T \phi(\mathbf{q})$. Let $\phi(\mathbf{p}) = [\phi_1(\mathbf{p}), \phi_2(\mathbf{p}), \phi_3(\mathbf{p}), \phi_4(\mathbf{p})]^T$. Then $\phi(\mathbf{p})^T \phi(\mathbf{q}) = \phi_1(\mathbf{p})\phi_1(\mathbf{q}) + \phi_2(\mathbf{p})\phi_2(\mathbf{q}) + \phi_3(\mathbf{p})\phi_3(\mathbf{q}) + \phi_4(\mathbf{p})\phi_4(\mathbf{q})$. Comparing the terms, we can identify the components of $\phi(\mathbf{p})$:

- $\phi_1(\mathbf{p})\phi_1(\mathbf{q}) = p_1^3 q_1^3 \implies \phi_1(\mathbf{p}) = p_1^3$
- $\phi_2(\mathbf{p})\phi_2(\mathbf{q}) = 3p_1^2 p_2 q_1^2 q_2 \implies \phi_2(\mathbf{p}) = \sqrt{3}p_1^2 p_2$
- $\phi_3(\mathbf{p})\phi_3(\mathbf{q}) = 3p_1 p_2^2 q_1 q_2^2 \implies \phi_3(\mathbf{p}) = \sqrt{3}p_1 p_2^2$
- $\phi_4(\mathbf{p})\phi_4(\mathbf{q}) = p_2^3 q_2^3 \implies \phi_4(\mathbf{p}) = p_2^3$

So, the feature mapping $\phi(\mathbf{p})$ can be defined as:

$$\phi(\mathbf{p}) = \begin{bmatrix} p_1^3 \\ \sqrt{3}p_1^2p_2 \\ \sqrt{3}p_1p_2^2 \\ p_2^3 \end{bmatrix}$$

Note that other forms are possible by multiplying components by -1 (e.g., $\phi_2(\mathbf{p}) = -\sqrt{3}p_1^2p_2$ and $\phi_2(\mathbf{q}) = -\sqrt{3}q_1^2q_2$), but this is a standard representation.

3.3 Question-3

1. Define $z_i = Lx_i \in \mathbb{R}^m$. Then

$$d_L(x_i, x_j) = \|z_i - z_j\|^2 = \|L(x_i - x_j)\|^2 = (x_i - x_j)^\top L^\top L (x_i - x_j).$$

Considering only class A with index set $\mathcal{S}_A$, we minimize

$$\min_L \sum_{i,j \in \mathcal{S}_A} \|L(x_i - x_j)\|^2,$$

where $x_i \in \mathbb{R}^n$, $L \in \mathbb{R}^{m \times n}$, so the objective is scalar.

2. If $L = \mathbf{0}$, then $d_L(x_i, x_j) = 0$ for all pairs and the objective is trivially minimized, but there is no discrimination. To avoid this, impose e.g. the constraint

$$\|L\|_F^2 = 1 \quad \text{or} \quad \text{trace}(L^\top L) = 1 \quad \text{or} \quad L^\top L = I_n,$$

or add a penalty $\lambda \|L\|_F^2$.

3. A Mahalanobis distance

$$d_A(x_i, x_j) = (x_i - x_j)^\top A (x_i - x_j)$$

matches

$$d_L(x_i, x_j) = \|L(x_i - x_j)\|^2 = (x_i - x_j)^\top (L^\top L) (x_i - x_j),$$

hence

$$\boxed{A = L^\top L.}$$

4. To enforce a margin $m > 0$ between class A and B:

$$\max_{L, m} m \quad \text{s.t.} \quad d_L(x_i, x_k) - \max_{p, q \in \mathcal{S}_A} d_L(x_p, x_q) \geq m, \quad \forall i \in \mathcal{S}_A, k \in \mathcal{S}_B, \quad \|L\|_F^2 = 1.$$

Alternatively, trade off tightness of A vs. separation from B by

$$\min_L \sum_{i,j \in \mathcal{S}_A} d_L(x_i, x_j) - \lambda \sum_{i \in \mathcal{S}_A, k \in \mathcal{S}_B} d_L(x_i, x_k) \quad \text{s.t.} \quad \|L\|_F^2 = 1.$$

Table 1: Marking Rubric for Question 3

Part	Criterion	Marks
(a)	Correct expression of transformed distance d_L	2
	Clear minimization objective over class A pairs	2
	Dimensionality check (scalar objective, matrix sizes consistent)	1
(b)	Reasoning of collapse when $L = 0$	2
	Valid constraint or regularization suggestion	3
(c)	Correct relation $A = L^\top L$	3
	Brief explanation why this factorization recovers the metric	2
(d)	Formulation of a margin-based objective or equivalent constraint	3
	Clear inclusion of both the separation objective and constraint	2

3.4 Question-4